4

In this chapter, we will learn about

- system of objects
- addition and resolution of forces
- forces in a plane and Newton's laws of motion

Force and motion (II)

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4.1 System of objects

1 Objects moving as a whole

Flipped classroom

Watch a video and answer the questions.

DSE goal

Apply Newton's second law of motion to solve problems.

DSE exam

12(1A)Q8, 14(1A)Q8, 17(1A)Q10, 19(1A)Q12, 21(1B)Q4, 22(1A)Q6, 23(1A)Q5, 23(1A)Q9

Sometimes, two or more objects may move together when we apply a force to only one object (Fig 4.1a). In such cases, do we need to apply newton's laws to each object separately if we want to study their motion?

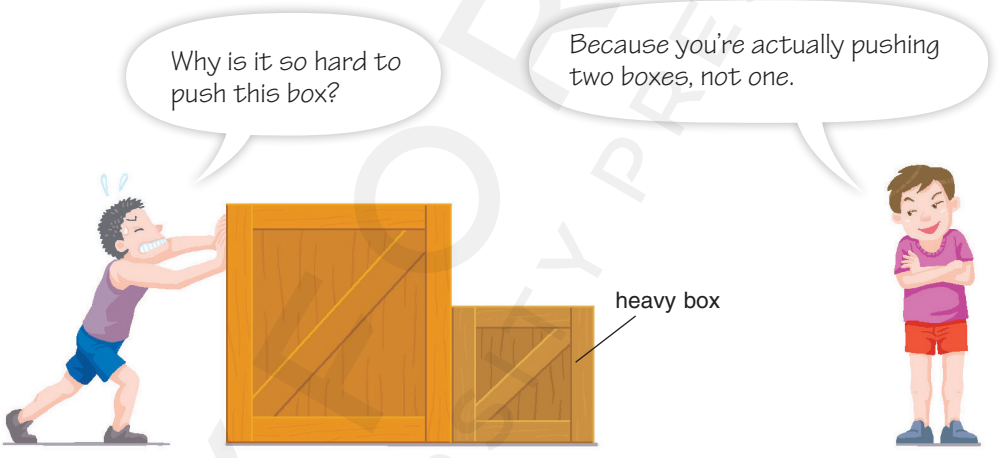


Fig 4.1a *Apply a force to one object to make two objects move.*

Consider the case when a man is pushing boxes *A* and *B* together on a smooth surface. Since the two boxes are moving as a whole (with **the same velocity and the same acceleration**), we may treat the combination of these two boxes as a **system** of objects when studying their motion (Fig 4.1b).

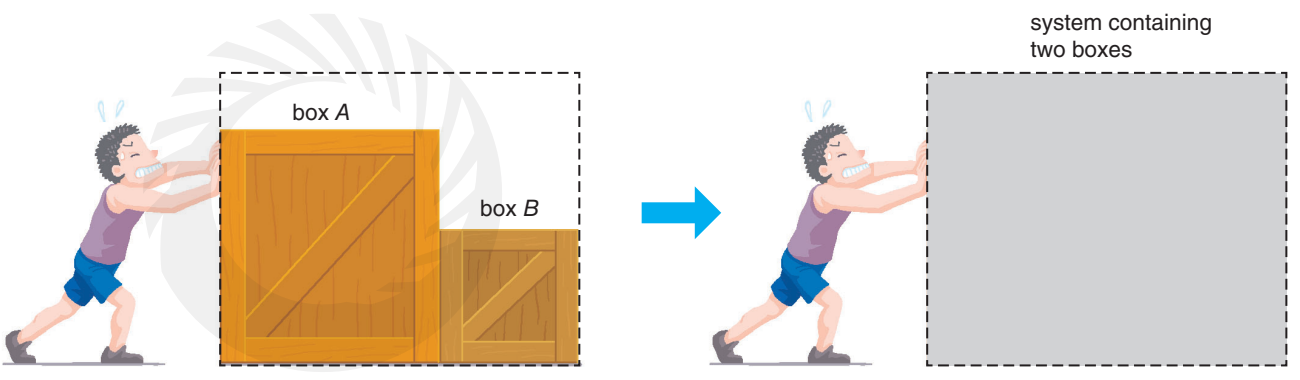


Fig 4.1b *Treat two boxes as a system of objects.*

In the above case, *A* and *B* act forces on each other. These are the **internal forces**, i.e. the forces between objects in the system. When treating the objects as one system, we can neglect all internal forces and consider only the **external forces**, i.e. the forces acting from outside the system (Table 4.1a).

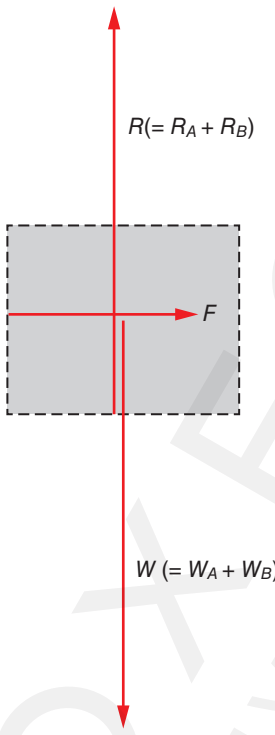
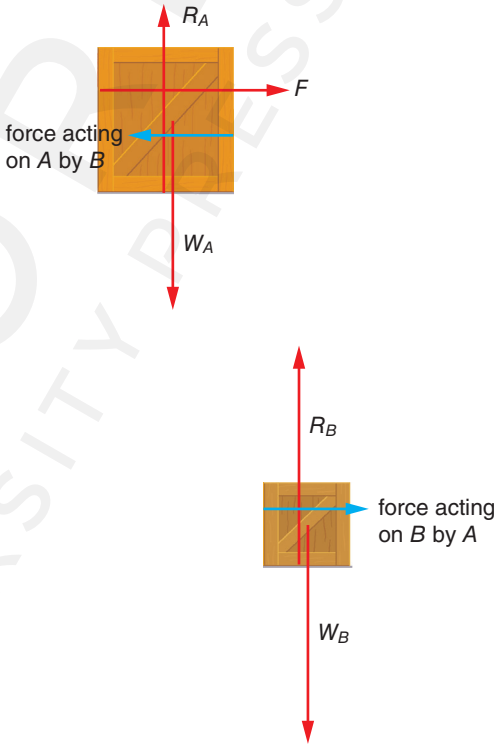
Treat boxes as a system of objects	Treat boxes as two objects
	

Table 4.1a *Internal forces (blue arrows) are usually neglected when objects are treated as a system.*

Note that the two internal forces in the above system form an action-and-reaction pair. They have the same magnitude but act along different directions.

Objects moving as a whole (with the same velocity and the same acceleration) can be treated as a system of objects. When we study their motion, we can neglect the internal forces of the system.

Example 1 Pushing two boxes

The masses of boxes A and B are 1 kg and 3 kg respectively. They are placed together on a smooth horizontal surface. A force of 4 N pushes A towards the right as shown (Fig a). Find the force acting on B by A.

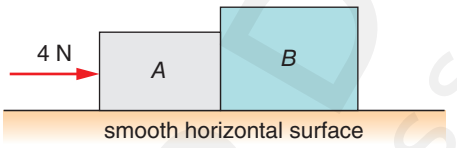


Fig a

Solution

Method 1: using the idea of system

Take the direction to the right as positive.

Consider the system of A and B (Fig b).

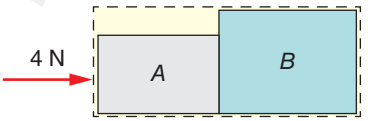


Fig b

By $F_{net} = ma$,

$4 = (1 + 3) \times a$

$a = 1 \text{ m s}^{-2}$

Consider B (Fig c). Let P be the magnitude of the force acting on B by A.

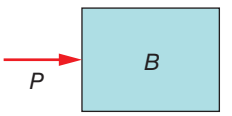


Fig c

By $F_{net} = ma$,

$P = 3 \times 1 = 3$ N

The force acting on B by A is 3 N (towards right).

Method 2: not using the idea of system

Take the direction to the right as positive.

Consider A (Fig d). Let P be the magnitude of the force acting on on A by B.

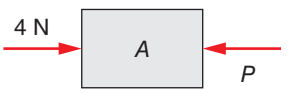


Fig d

By $F_{net} = ma$,

$4 + (-P) = 1a$

$a = 4 - P$ ..... (1)

Consider B (Fig e).

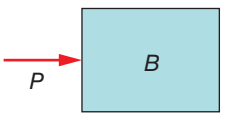


Fig e

By $F_{net} = ma$,

$P = 3a$ ..... (2)

Substitute (1) into (2),

$P = 3 \times (4 - P)$

$P = 12 - 3P$

$P = 3$ N

The force acting on B by A is 3 N (towards right).

We can see that the answer can be obtained by using either method. The calculation involved is usually simpler when we work out the acceleration of an object using the idea of system. Choosing an appropriate method can help us quickly solve a problem and avoid calculation mistakes. ▶

If the frictions between the two boxes are not large enough, the boxes will not move as a whole.

Consider the case when two stacked boxes are moved by a man on a rough surface (Fig 4.1c). If the frictions between the two boxes are large enough, the boxes will move together with **the same velocity and the same acceleration**. We may then treat these stacked objects as a system of objects.

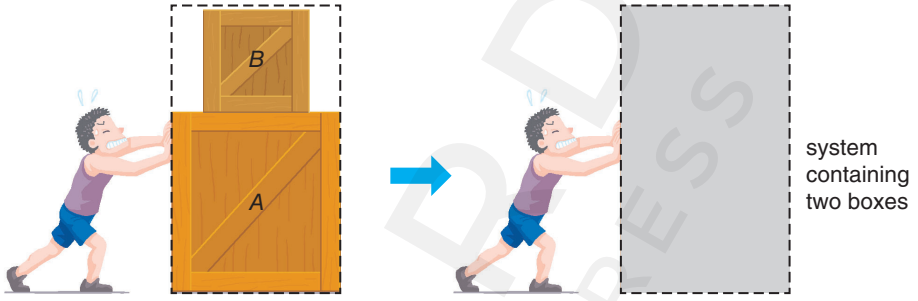


Fig 4.1c *Treating two stacked boxes as a system.*

We can see that a friction does not always stop or slow down an object.

In this case, the frictions between the two boxes are internal forces of this system (Table 4.1b). These two frictions form an action-and-reaction pair—they have the same magnitude but act along opposite directions. Note that it is the friction acting on B that makes B accelerate.

Right and wrong

A lift car is connected to a counterweight via a cable wound on a drum. A motor is used to turn the drum to drive the lift car (Fig a).

If the friction between the drum and the cable disappears,

- ✗ it would be easier for the drum to drive the lift car.
- ✓ the drum cannot drive the lift car at all.

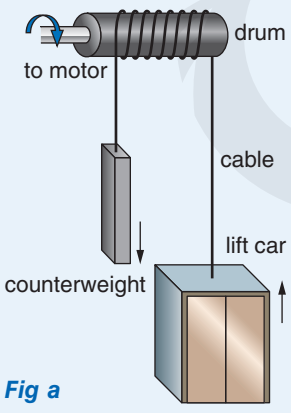


Fig a

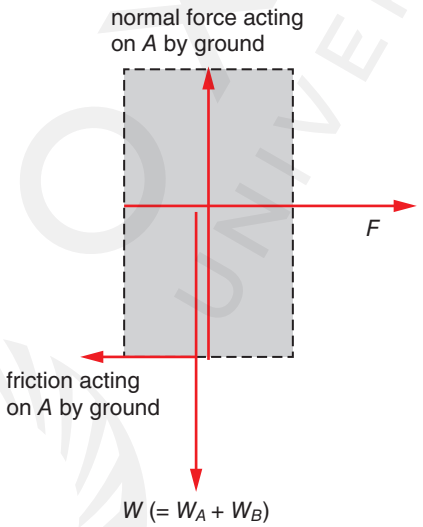
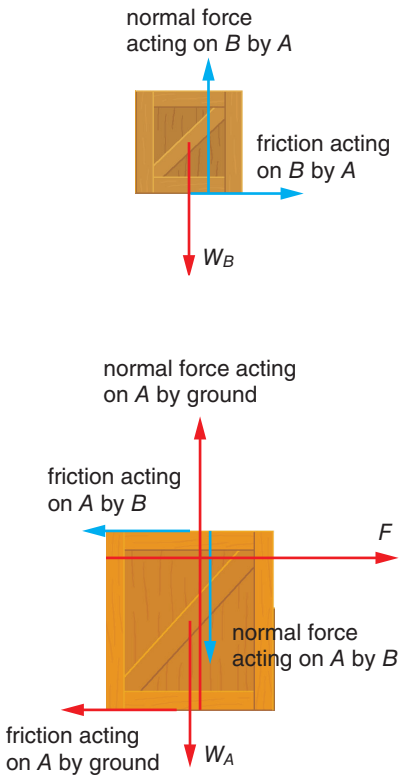
Treat boxes as a system	Treat boxes as two objects
	

Table 4.1b *Identifying internal forces (blue arrows) acting on a system of stacked objects.*

Example 2 | Stacked objects

An apple (A) and a book (B) are initially at rest on a rough horizontal table. The masses of A and B are 0.5 kg and 0.8 kg respectively. A horizontal force of 14 N is applied to B (Fig a) and they start to move to the right. They move together for 3 m after 2 s.

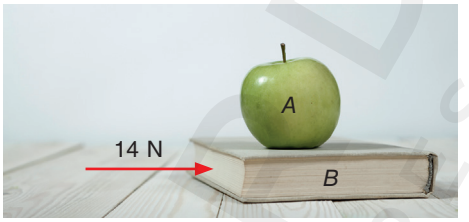


Fig a

(a) Find the acceleration of A and B.

(b) Find the friction acting on A by B.

(c) Find the friction acting on B by the table.

Solution

Take the direction to the right as positive.

(a) Consider the system of A and B.

$u = 0$ m s^{-1}, $s = +3$ m, $t = 2$ s, $a = ?$

By $s = ut + \frac{1}{2}at^2$,

$$3 = 0 \times 2 + \frac{1}{2} \times a \times 2^2$$

$$a = 1.5 \text{ m s}^{-2}$$

The acceleration of A and B is 1.5 m s^{-2} (towards right).

(b) Consider A (Fig b).

$F_{net} = ma$

$f_{AB} = 0.5 \times 1.5$

$= 0.75$ N

The friction acting on A by B is 0.75 N (towards right).

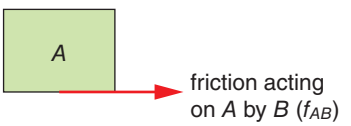


Fig b

(c) Consider the system of A and B (Fig c).

$F_{net} = ma$

$14 + (-f) = (0.5 + 0.8) \times 1.5$

$f = 12.1$ N

The friction acting on B by the table is 12.1 N (towards left).

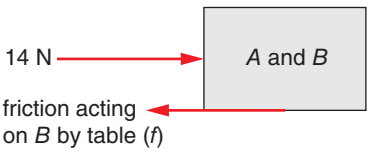


Fig c

Quick check 1

→ E-version in OUP Exercise Platform is also available.

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(For Q1–2.) A girl is holding four boxes *W* to *Z* as shown (Fig a).

Fig a

1 Consider the four boxes as a system. Which of the following is/are the external force(s) of the system? Put a tick in the corresponding box(es).

- ☐ The upward force provided by the girl on *W*
- ☐ The upward force provided by *W* on *X*
- ☐ The downward force provided by *Z* on *Y*
- ☐ The total weight of the four boxes

2 Which of the following is/are the force(s) acting on *X*? Put a tick in the corresponding box(es).

- ☐ The upward force provided by the girl on *W*
- ☐ The upward force provided by *W* on *X*
- ☐ The downward force provided by *Y* on *X*
- ☐ The weight of *X*

(For Q3–4.) Blocks *A* and *B* are stacked on a rough horizontal surface as shown (Fig b). *B* is pushed by a horizontal force *F* and the two blocks move to the left together.

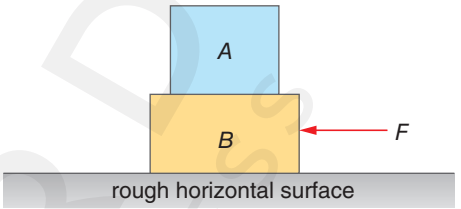


Fig b

3 Which of the following forces point(s) to the left? Put a tick in the corresponding box(es).

- ☐ The friction acting on *B* by the horizontal surface
- ☐ The friction acting on *B* by *A*
- ☐ The friction acting on *A* by *B*

4 Consider *A* and *B* as a system. Which of the following is/are the external force(s) of the system? Put a tick in the corresponding box(es).

- ☐ The pushing force *F*
- ☐ The friction acting on *A* by *B*
- ☐ The friction acting on *B* by the horizontal surface
- ☐ The normal force acting on *B* by the horizontal surface
- ☐ The normal force acting on *A* by *B*
- ☐ The total weight of *A* and *B*

Learn by practice 1

1 Two 4-kg blocks *X* and *Y* are placed on a smooth horizontal surface (Fig a). *X* is pushed by a horizontal force of 16 N so that the two blocks move towards the right together.

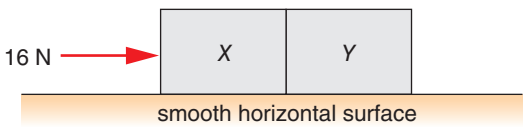


Fig a

(a) Find the acceleration of the two blocks.

(b) Find the force acting on *X* by *Y*.

2 Tension and connected objects

Imagine a child pulls two toys together on a smooth surface (Fig 4.1d). If string *X* connecting the toys is light, inextensible and remains taut, the toys will move with **the same velocity and the same acceleration**. Again, we may treat these connected objects as a system of objects (Fig 4.1e).

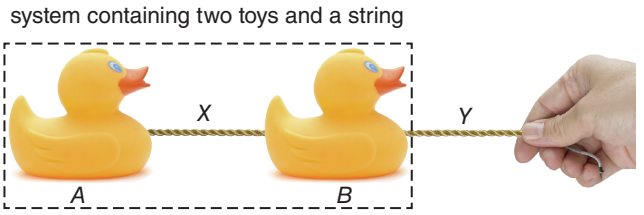


Fig 4.1d *Treating two connected objects as a system.*

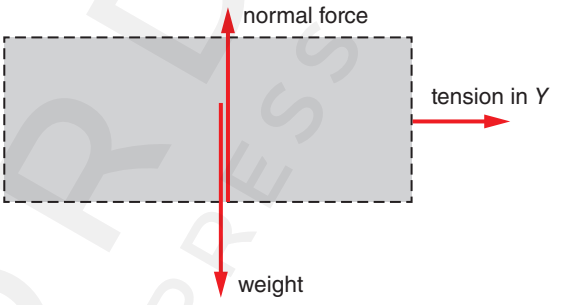


Fig 4.1e *The free-body diagram of the system.*

Bear in mind that the tension in string *X* acting on *A* and that acting on *B* are internal forces when *A* and *B* are treated as objects in the same system.

Example 3 | Objects connected by a string

The masses of boxes *X* and *Y* are 2 kg and 3 kg respectively. They are connected by a light inextensible string and placed on a smooth horizontal surface. *X* is pulled by a 10-N force to the left as shown (Fig a). Taking the direction to the left as positive, find the acceleration a of *X* and the magnitude of tension T in the string.

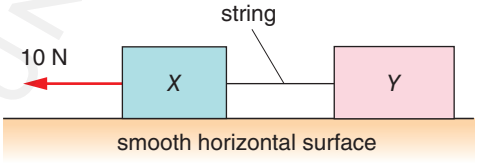


Fig a

Solution

Consider the system of *X* and *Y* (Fig b).

By $F_{net} = ma$,

$10 = (2 + 3) \times a$

$a = 2 \text{ m s}^{-2}$

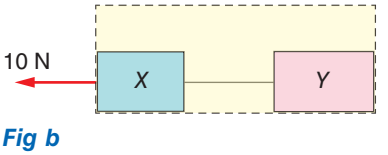


Fig b

Consider *Y* (Fig c).

By $F_{net} = ma$,

$T = 3 \times 2 = 6 \text{ N}$

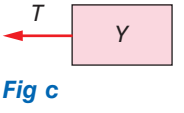
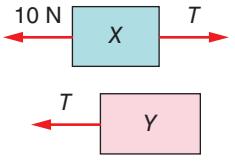


Fig c

We can also solve this problem without using the idea of system:

1 Set up two equations by considering *X* and *Y* separately.

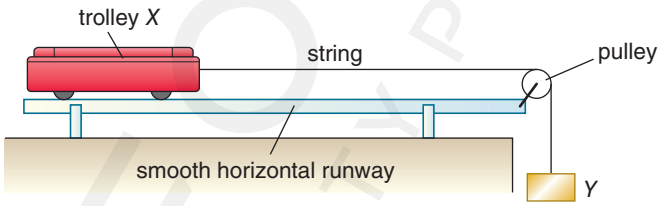


2 Solve the two equations to find *T*.

Sometimes, two objects are connected by a light, inextensible string that passes over a smooth pulley. As long as the string remains taut, the velocities of the objects will have the same magnitude, and so will the accelerations. The following example shows how a problem involving pulleys can be solved.

Example 4 Objects connected by a string over a pulley

On a smooth horizontal runway, a 0.8-kg trolley X is connected to a 0.1-kg object Y with a light, inextensible string over a smooth pulley. When Y is released, it moves downwards with an acceleration a.

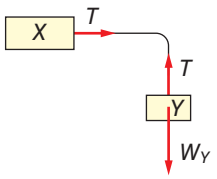


(a) Find a.

(b) Hence, find the magnitude of tension T in the string.

Solution

(a) Consider the system of X and Y.
Consider the external force(s) along the directions of motion.



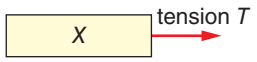
By $F_{net} = ma$,

$$W_Y = (m_X + m_Y)a$$

$$0.1 \times 9.81 = (0.8 + 0.1) \times a$$

$$a = 1.09 \text{ m s}^{-2}$$

(b) Consider X.



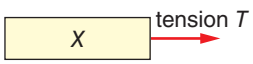
By $F_{net} = ma$,

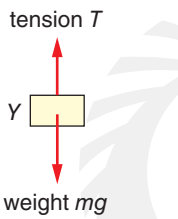
$$T = 0.8 \times 1.09$$

$$= 0.872 \text{ N}$$

We can also solve this problem without using the idea of system:

1 Set up two equations by considering X and Y separately.





2 Solve the two equations to find a and T.

Think about...

If this experiment were done on the Moon where the acceleration due to gravity is smaller than that on the Earth, would the value of a be larger or smaller than the answer in (a)? Smaller

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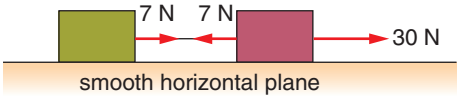
Quick check 2

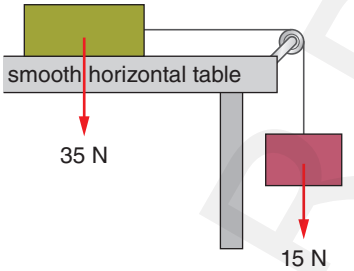
→ E-version in OUP Exercise Platform is also available.

1 In each of the following cases, consider the two blocks connected by a string as a system. The equation $F_{net} = ma$ can be applied to find the acceleration of the blocks where m is the total mass of the system. Find the value taken as F_{net} in each case.

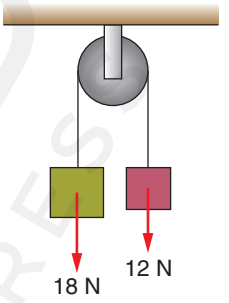
(a)



(b)



(c)



2 Trolley P is connected to object Q of mass m with a string over a smooth pulley (Fig a). When they are released, the tension in the string is T.

Determine whether each of the following statements is correct.

(a) The net force acting on P is T.

(b) The net force acting on Q is mg.

(c) P and Q have the same magnitude of acceleration.

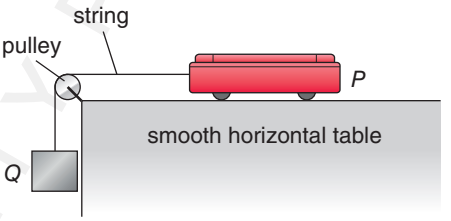


Fig a

Learn by practice 2

1 The masses of blocks A and B are 6 kg and 4 kg respectively. They are connected by a string as shown (Fig a).

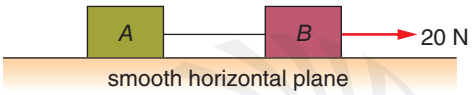


Fig a

If A is pulled by a horizontal force of 20 N and the two blocks move to the right together, what is the tension in the string?

2 Blocks X and Y are connected with a string over a smooth pulley as shown (Fig b). Each block has a mass of 2 kg.

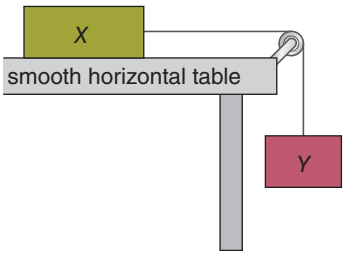


Fig b

The blocks are now released from rest. Find the tension in the string.

Exercise 4.1

(For Q1–2.) Boxes X and Y, each of mass 0.5 kg, are stacked up on a horizontal table (Fig a). The maximum friction between X and the table is 5 N and that between X and Y is very large. Both boxes are at rest.

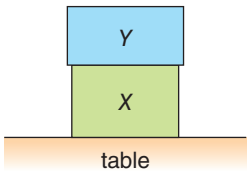


Fig a

★**1** If a horizontal force of 4 N acts on X, what will be the friction between X and Y?

A 0
B 1 N
C 2 N
D 4 N

★**2** If a horizontal force of 9 N acts on X, what will be the friction between X and Y?

A 0
B 1 N
C 2 N
D 4 N

3 The masses of blocks P and Q are 6 kg and 2 kg respectively. They are placed on a smooth horizontal surface as shown (Fig b). Q is pushed by a horizontal force of 12 N and the two blocks move to the left together. What is the magnitude of the contact force between P and Q?

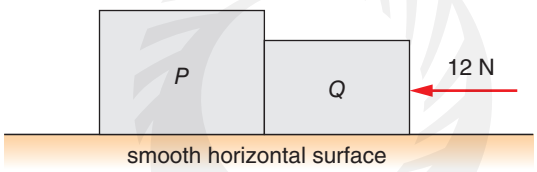


Fig b

4 A trolley of mass 2 kg is connected to a block by a string over a smooth pulley (Fig c). At first, the trolley is held stationary on a frictionless horizontal table.

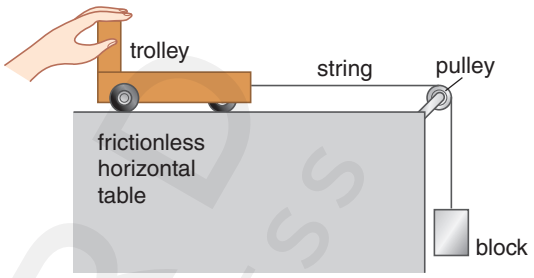


Fig c

When the trolley is released, the tension in the string is 4 N.

(a) What is the acceleration of the block?

(b) What is the mass of the block?

★**5** The masses of blocks P and Q are 2 kg and 3 kg respectively. They are hung over a smooth pulley as shown (Fig d). The blocks are now released from rest.

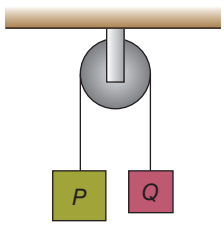


Fig d

(a) Find the acceleration of Q.

(b) Find the tension in the string.

★**6** Three blocks P, Q and R, each of mass 2 kg, are connected by two strings A and B on a smooth horizontal plane (Fig e). A horizontal force F of 30 N pulls P towards the right and the blocks move to the right together. The tensions in A and B are T_1 and T_2 respectively.

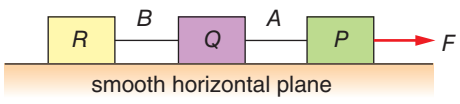


Fig e

(a) Find the net force acting on P.

(b) Which force, F, T_1 or T_2, has the smallest magnitude?

4.2 Addition and resolution of forces

1 Adding forces using vector diagram

DSE goal

Find the vector sum of coplanar forces graphically and algebraically.

DSE exam

12(1A)Q5, 13(1A)Q6

When an object is acted on by different forces, its direction of acceleration depends on the net force, instead of any individual force (Fig 4.2a). However, how can we determine the net force when the individual forces have different directions?

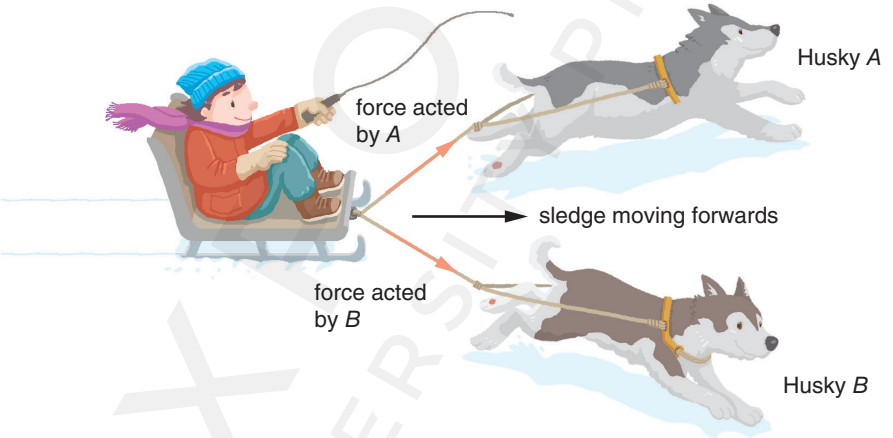


Fig 4.2a *Two huskies pulling a sledge along different directions.*

→ This simulation shows how two forces can be added using (a) the 'tip-to-tail' method, and (b) the parallelogram of forces method.

Simulation

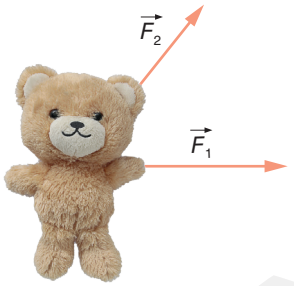


Fig 4.2b *Two forces $\vec{F}_1$ and $\vec{F}_2$ acting on an object.*

Forces are vectors and can be added directly in a vector diagram. Imagine there are two force vectors $\vec{F}_1$ and $\vec{F}_2$ acting on an object along different directions (Fig 4.2b). Table 4.2a shows two methods of finding the vector sum of the forces, namely *parallelogram of force method* and *tip-to-tail method*.

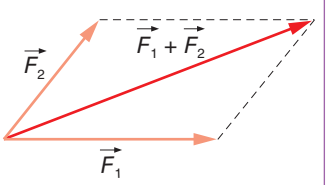
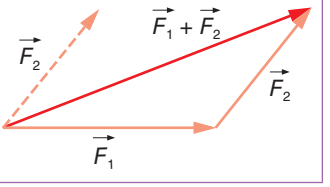
Parallelogram of forces method	
① Draw a parallelogram so that: • $\vec{F}_1$ and $\vec{F}_2$ are adjacent sides; • the tails of $\vec{F}_1$ and $\vec{F}_2$ meet at one point. ② Arrow pointing diagonally from the tails of $\vec{F}_1$ and $\vec{F}_2$ represents the resultant force.	
Tip-to-tail method	
① Shift $\vec{F}_2$ so that its tail is at the tip of $\vec{F}_1$. ② Arrow from the tail of $\vec{F}_1$ to the tip of $\vec{F}_2$ represents the resultant force.	

Table 4.2a *Finding the vector sum of two forces $\vec{F}_1$ and $\vec{F}_2$ in a vector diagram.*

Skill *Drawing*

Vector diagram

In a vector diagram, vectors having the same length and direction are considered as equal, regardless of where they are placed. With this property, we can shift vectors in the diagram.

→ Teaching notes and exercises are available on OUP web.

parallelogram of force method 力的平行四邊形法則　　tip-to-tail method 首尾連接法

When the angle between two forces increases, the magnitude of the resultant force decreases (refer to DSE 2012 Paper 1A Q5). Ask Ss to do Exercise 4.2 Q2 on p.170 to verify this using the graphical method.

If the forces are drawn in scale with the correct orientation, the magnitude and direction of the resultant force can then be found by direct measurement. Besides this, they can also be found by calculation, as shown in the following example.

→ This is a Video Quiz about the experiment 'Addition of forces'.

Example 5 Tip-to-tail method

Danny and Frankie exert pushing forces of 30 N and 40 N respectively on a table. The two forces are perpendicular to each other (Fig a).

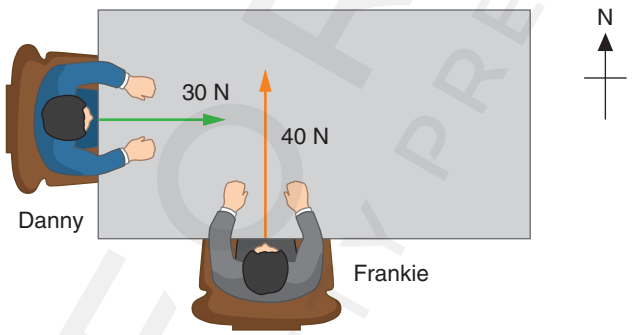


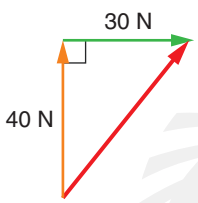
Fig a

(a) Draw the two forces and their resultant force in a new diagram.

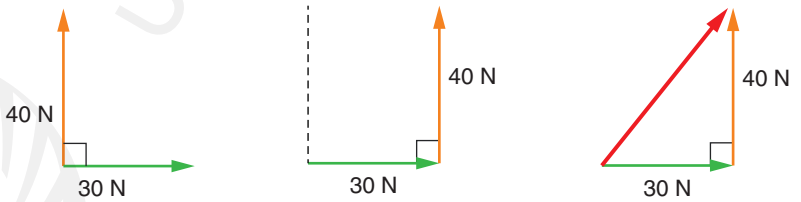
(b) Find the direction and magnitude of the resultant force in (a) by calculation.

Solution

If we shift the 30-N vector instead of the 40-N one, the same answer will be obtained.



(a) ① Draw the two forces. ② Shift one of the forces so that the two forces are 'tip to tail'. ③ Draw an arrow from the tail to the tip. This is the resultant force.



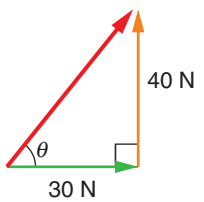
(b) Magnitude $= \sqrt{30^2 + 40^2} = 50$ N

Direction:

$\tan\theta = \frac{40}{30}$

$\theta = 53.1°$

The resultant force is 50 N (N 36.9° E).



When there are more than two forces, we may repeatedly apply the parallelogram method or the tip-to-tail method to two forces every time until the resultant force is found. We can also find the resultant force in one go by using the tip-to-tail method. See the following example.

Example 6 Net force acting on a sledge

Three huskies pull a sledge towards the left (Fig a). Figure b shows the tensions $\vec{P}$, $\vec{Q}$, $\vec{R}$ and friction $\vec{f}$ acting on the sledge. Find graphically the resultant of all these forces.

Fig a

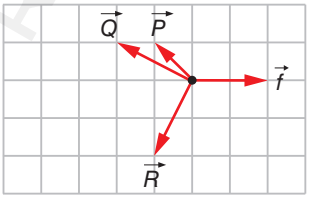
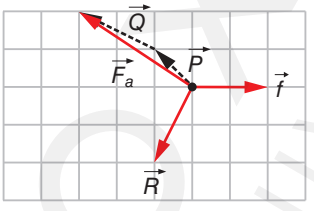


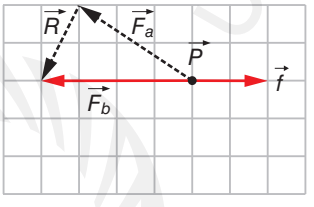
Fig b

Solution

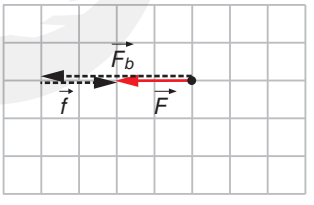
① Adding $\vec{P}$ and $\vec{Q}$ gives $\vec{F}_a$.



② Adding $\vec{F}_a$ and $\vec{R}$ gives $\vec{F}_b$.

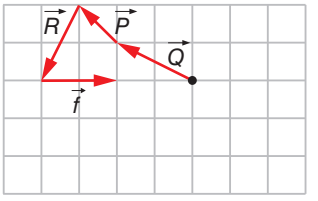


③ Adding $\vec{F}_b$ and $\vec{f}$ gives the resultant force $\vec{F}$.

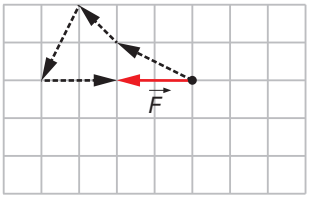


Alternative method:

① Shift the forces to make them 'tip to tail'.

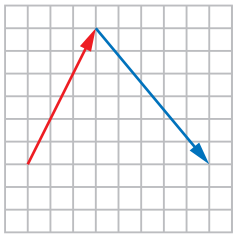


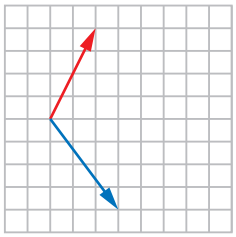
② Draw an arrow from the tail to the tip. This is the resultant force $\vec{F}$.

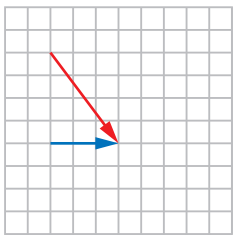


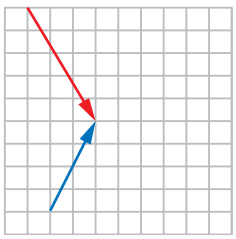
Learn by Practice 3

1 Draw the resultant force of the two forces in each of the following cases.

(a) 

(b) 

(c) 

(d) 

2 A lamp is hung on a string (Fig a). The tension in the string is 5 N. Find the magnitude of the resultant force acting on the lamp by the string.

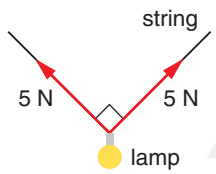


Fig a

3 Two forces act on a block as shown (Fig b). Find the resultant force.

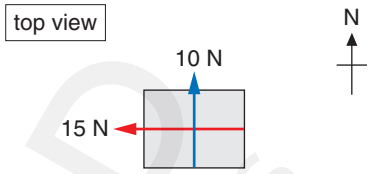


Fig b

4 The diagram shows two forces (Fig c).

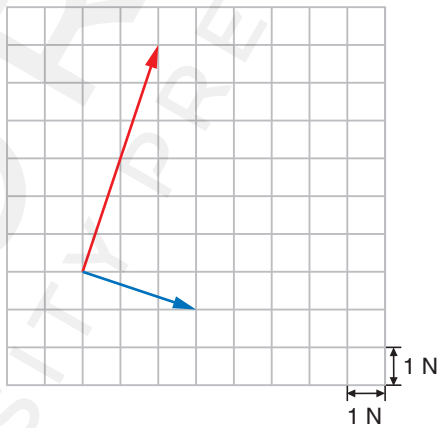


Fig c

(a) Draw the resultant force in the diagram.

(b) Hence, find the magnitude of the resultant force.

DSE goal

Resolve a force graphically and algebraically into components along two mutually perpendicular directions.

Simulation

→ This simulation shows how to resolve a force into two perpendicular components.

2 Resolving forces into components

We can add two forces to get a resultant force. Conversely, we can **resolve** a force $\vec{F}$ into two forces $\vec{F_1}$ and $\vec{F_2}$ whose total effect is the same as that of $\vec{F}$. The resolved forces $\vec{F_1}$ and $\vec{F_2}$ are called the **components** of $\vec{F}$. Usually we choose a pair of components that are **perpendicular** to each other (Fig 4.2c).

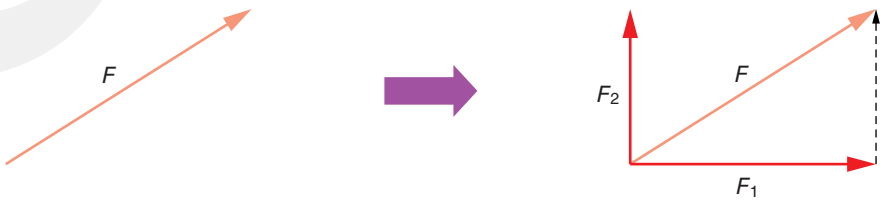


Fig 4.2c *Resolving a force into two perpendicular components.*

resolve 分解　component 分量

In Figure 4.2d, a 5-N force is resolved into two perpendicular components, one of which makes an angle of 30° with the 5-N force. The magnitudes of the components can be calculated using trigonometric functions.

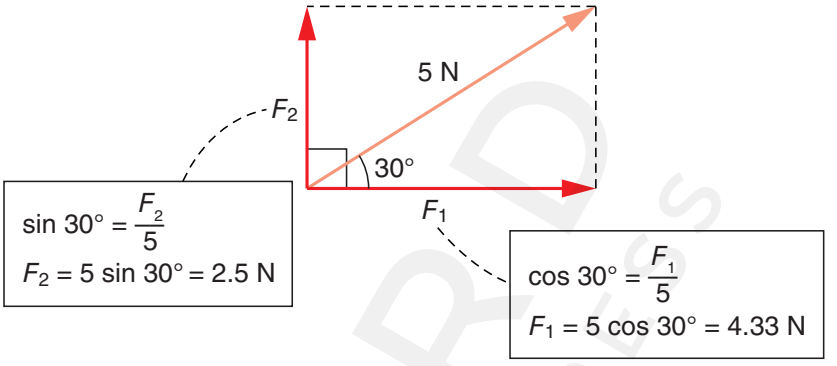
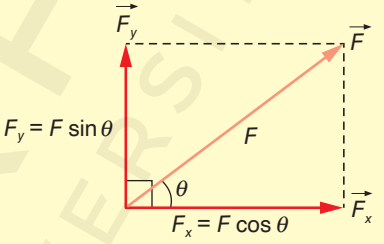


Fig 4.2d *Calculating the magnitudes of the components.*

Remind Ss that if a force is resolved into components that are not perpendicular to each other, it is hard to find their magnitudes.

In general, if a force $\vec{F}$ is resolved into perpendicular components $\vec{F_x}$ and $\vec{F_y}$, with $\vec{F_x}$ making an angle θ with $\vec{F}$, then

$$F_x = F \cos \theta \quad \text{and} \quad F_y = F \sin \theta$$

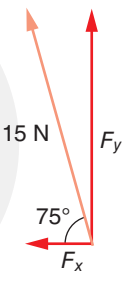
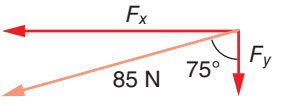
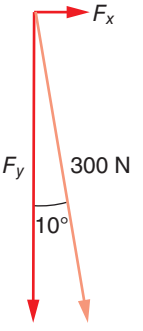


Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 Complete the table to resolve the given force into a horizontal component F_x and a vertical component F_y.

			
F_x			
F_y			

We can add forces that are not perpendicular to each other by first resolving the forces. Let us see how this can be done.

Example 7 Jumping frog

A frog of weight 8 N jumps from the ground. The force acting on its feet by the ground is 20 N and makes an angle of 40° to the horizontal (Fig a). Find the net force acting on the frog.

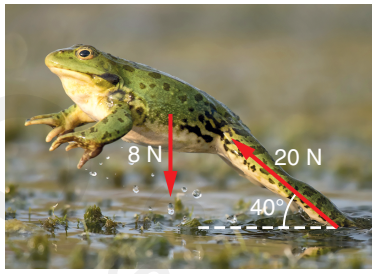


Fig a

Solution

$F_x = 20 \cos 40° = 15.3$ N

$F_y = 20 \sin 40° = 12.86$ N

As $F_y > W$, the vertical component of net force points upwards (Fig c).

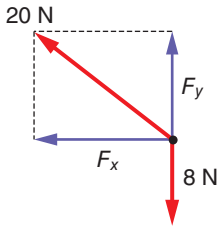


Fig b

Horizontal component of $F_{net} = F_x = 15.3$ N

Vertical component of $F_{net} = 12.86 - 8 = 4.86$ N

Magnitude:

$F = \sqrt{15.3^2 + 4.86^2} = 16.1$ N

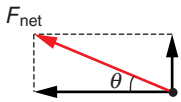


Fig c

Direction:

$$\tan\theta = \frac{4.86}{15.3}$$

$\theta = 17.6°$

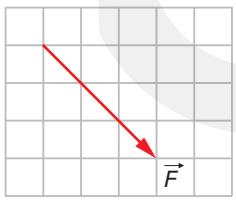
The net force is 16.1 N (to the left at 17.6° above the horizontal).

Tip
Add components along the same direction.

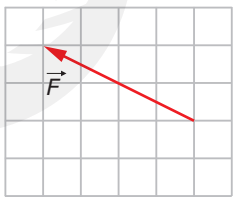
Learn by Practice 4

1 In each of the following cases, draw the components of force $\vec{F}$ along the horizontal and vertical directions.

(a)



(b)



2 A suitcase is pulled by a force of 10 N (Fig a). Find the horizontal and vertical components of the force.

Fig a

3 Two forces act on a particle (Fig b). Find the resultant force.

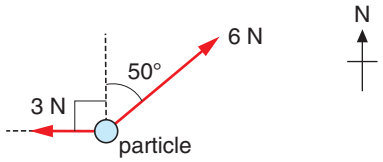
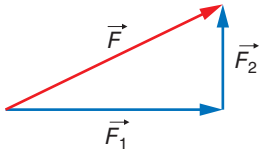
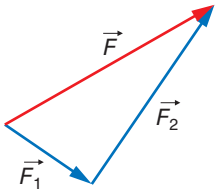


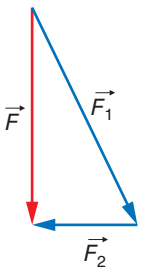
Fig b

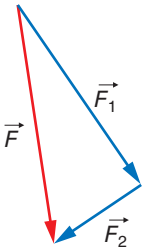
Exercise 4.2

1 Which of the following is **not** a correct way to resolve a force $\vec{F}$ into two perpendicular components $\vec{F}_1$ and $\vec{F}_2$?

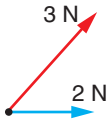
A 

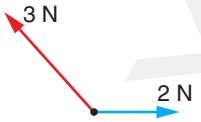
B 

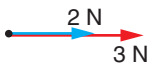
C 

D 

***2** A 2-N force and a 3-N force act on the same point in different ways as shown. Arrange the four cases in ascending order of the magnitudes of the resultant forces.

(1) 

(2) 

(3) 

(4)

A (1), (3), (2), (4)

B (1), (4), (2), (3)

C (4), (1), (2), (3)

D (4), (2), (1), (3)

3 A picture is hung by a string as shown (Fig a). The tension in the string is 6 N. Find the resultant of the forces acting on the picture by the string.

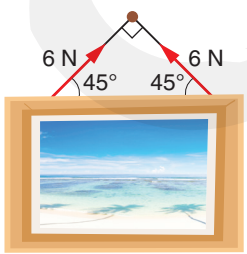


Fig a

4 A girl sits on a rope swing (Fig b). The rope makes an angle of 60° to the horizontal. The tension in the rope is T and its horizontal component is 200 N. Find T.

Fig b

***5** Object O is at the centre of a right octagon. Eight forces act on it as shown (Fig c).

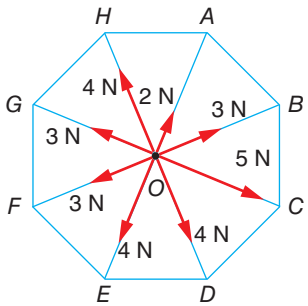
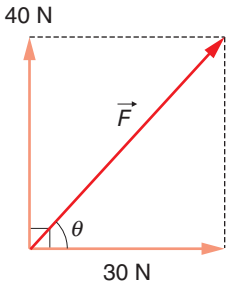


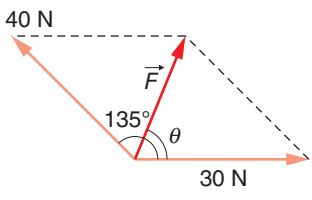
Fig c

(a) Find the resultant of the forces.

(b) Suppose one more force is added such that the net force acting on O becomes zero. What is the magnitude and direction of this force?

***6** In each of the following cases, $\vec{F}$ is the resultant of the 30-N force and 40-N force. Find its magnitude F and the value of θ.

(a) 

(b) 

4.3 Forces in a plane and Newton's laws of motion

1 Objects at rest or in uniform motion

DSE goal

Apply Newton's second law of motion to solve problems.

DSE exam

13(1A)Q5, 14(1B)Q4, 15(1B)Q4, 15(1B)Q5, 18(1A)Q5, 18(1A)Q8, 20(1B)Q5, 23(1A)Q6, 23(1A)Q7, etc

When an object remains at rest or in uniform motion, by Newton's first law of motion, the net force acting on it must be zero. In other words, the resultant force acting on the object along **any direction** is zero. Let us see how this can be applied when we solve problems in real-life scenarios.

Example 8 Sign board hung by ropes

A stationary sign board of weight 60 N is hung by two ropes R_1 and R_2 as shown (Fig a). The tensions in R_1 and R_2 are T_1 and T_2 respectively. Find T_1 and T_2.

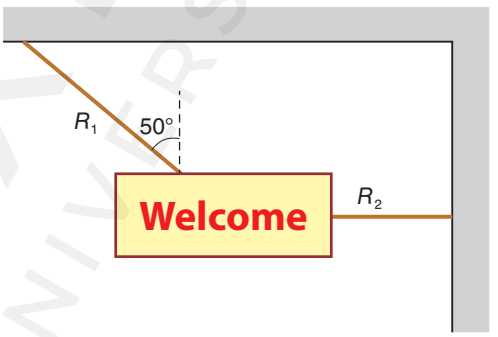


Fig a

Solution

Resolve T_1 as shown (Fig b).

Consider the vertical direction.

$T_1 \cos 50° = 60$

$T_1 = 93.3$ N

Consider the horizontal direction.

$T_2 = 93.3 \sin 50° = 71.5$ N

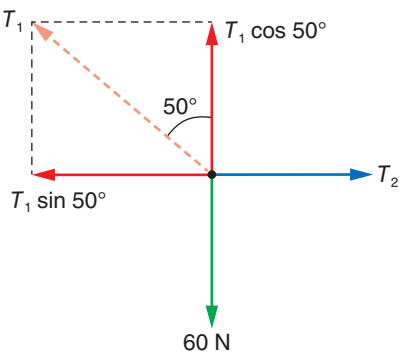


Fig b

Tip

Since the board is stationary, the resultant force along any direction is zero.

☆ HKDSE 2013 Paper 1A Q5

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Monkey hanging from two ropes

A monkey of weight 180 N hangs from two ropes and remains stationary as shown (Fig a). Find the value of tension T.

A 90 N
B 116 N
C 138 N
D 280 N

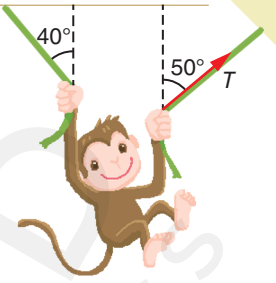


Fig a

Solution

According to the free-body diagram (Fig b), T and T_1 are perpendicular to each other. Therefore, we can resolve W along the directions of the tensions (Fig c).

We can also resolve the tensions along the vertical and horizontal directions. However, the calculation involved is more complicated and we may make mistakes more easily.

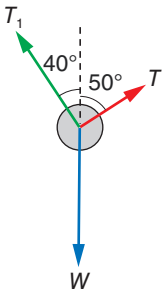


Fig b

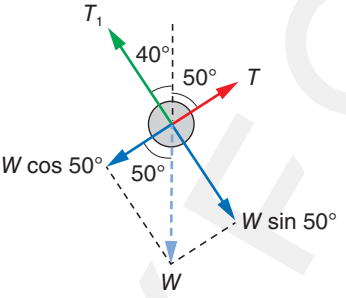


Fig c

$T = W \cos 50° = 180 \cos 50° = 116$ N

∴ The answer is B.

Revision 4 Q16 (p.186)

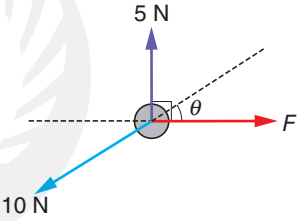
Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

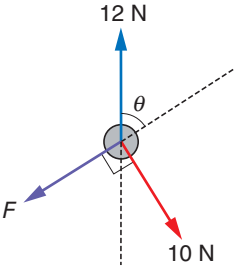
1 The following are the free-body diagrams of different stationary objects. In each case, determine which equation(s) is/are correct. Put a tick in the corresponding box(es).

(a)



- ☐ $F \cos \theta = 10$
- ☐ $F \sin \theta = 5$
- ☐ $F = 10 \cos \theta$
- ☐ $5 = 10 \sin \theta$

(b)



- ☐ $F \cos \theta = 12$
- ☐ $F \sin \theta = 12 \sin \theta$
- ☐ $F = 12 \cos \theta$
- ☐ $10 = 12 \sin \theta$

Example 9 Flying jet suit

A jet suit is a device that allows a person to fly by providing a thrust. The thrust is produced by a pair of arm engines and a back engine of the suit (Fig a).

Fig a

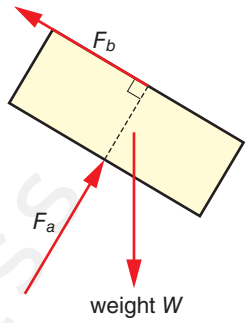


Fig b

A man in a jet suit flies at a constant velocity. His free-body diagram is shown in Figure b. F_a and F_b are the thrust provided by the arm engines and the back engine respectively. Find the total mass of the man and the flying jet suit if F_a = 765 N and F_b = 440 N.

Solution

Tip
Since the man is flying at a constant velocity, the net force is zero.

W = resultant force of F_a and F_b

$= \sqrt{765^2 + 440^2}$

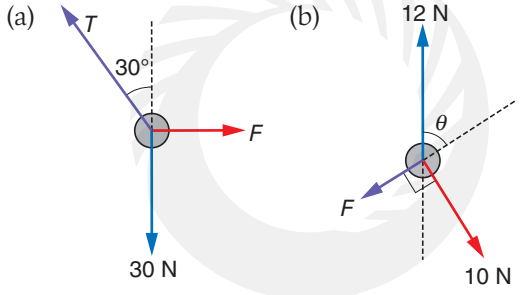
$= 883$ N

By $W = mg$,

$$m = \frac{W}{g} = \frac{883}{9.81} = 90.0 \text{ kg}$$

Learn by Practice 5

1 The following are the free-body diagrams of different objects moving in uniform motions. Find F in each case.



2 A puppy pulls a towel and remains stationary on the ground (Fig a). The towel makes an angle of 30° to the horizontal and the tension in it is 10 N. Find the friction acting on the puppy by the ground.

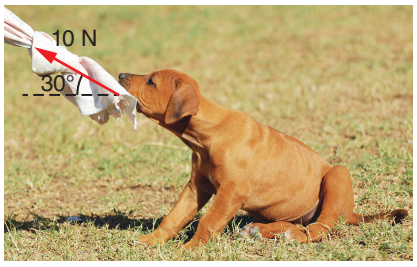


Fig a

2 Objects moving under a net force

Simulation

→ This simulation shows how to alter the acceleration of a box. It allows Ss to change the size and the angle of the applied force on the box. The weight of the box and the friction can also be changed. It is available on OUP web in PC version (.exe) only.

According to Newton's second law of motion, an object accelerates along the direction of the net force acting on it. When solving problems, we can resolve forces in a way that one of their components is along the moving direction of the object.

Skill *Problem-solving*

Resolving forces when net force is non-zero

① Draw a free-body diagram for the object of interest.

② Choose two perpendicular directions that are convenient to solve the problem, e.g. parallel and perpendicular to the object's moving direction.

③ Define the positive directions.

④ Resolve forces and find the resultant force in each direction in ②.

⑤ Apply the equation $F_{net} = ma$ and solve for the unknowns.

→ Teaching notes and exercises are available on OUP web.

Example 10 Acceleration of a toy truck

A toy truck is pulled by a girl with a string and accelerates horizontally to the right. The string is at an angle of 25° above the horizontal (Fig a). The tension in the string is 2.4 N. The friction acting on the truck by the ground is 2 N. The mass of the truck is 300 g.

Fig a

(a) Find the acceleration of the truck.

(b) Find the normal force acting on the truck by the ground.

Solution

(a)

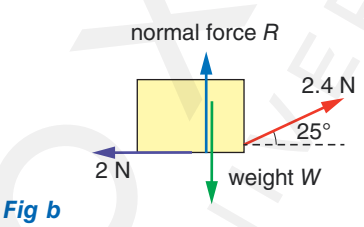


Fig b

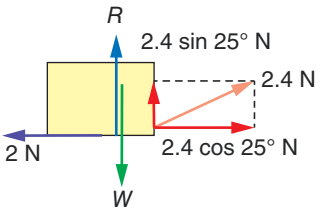


Fig c

Consider the horizontal direction. Take the direction towards the right as positive.

Net force = 2.4 cos 25° + (−2) = 0.1751 N

By $F_{net} = ma$,

$$a = \frac{F_{net}}{m} = \frac{0.1751}{0.3} = 0.584 \text{ m s}^{-2}$$

The acceleration is 0.584 m s^{-2} (towards right).

(b) Consider the vertical direction. Take upwards as positive.

$$F_{net} = 0$$
$$R + 2.4 \sin 25° + (-W) = 0$$
$$R + 2.4 \sin 25° - 0.3 \times 9.81 = 0$$
$$R = 1.93 \text{ N}$$

The normal force is 1.93 N (upwards).

Tip

Since the truck does not move in vertical direction, the resultant force along this direction is zero.

Example 11 | Acceleration of a hanging object

An air freshener is hung in a car (Fig a). When the car is stationary, the string hanging the air freshener remains vertical. When the car is speeding up, the string makes an angle θ to the vertical (Fig b). θ is constant if the acceleration of the car is constant.

Fig a

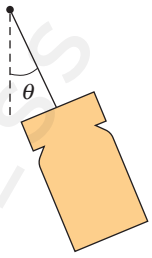


Fig b

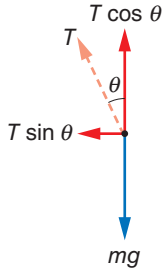
(a) Explain, in terms of inertia, why the string becomes inclined when the car is speeding up.

(b) Suppose the car accelerates to the right and θ becomes 10°. Find the magnitude of the car's acceleration.

Solution

(a) When the car is speeding up, the air freshener tends to move at its original velocity due to inertia. The speed of the air freshener becomes lower than the car's speed. As a result, the air freshener moves backwards relative to the car and the string becomes inclined.

(b) Consider the vertical direction.



$T \cos \theta = mg$ .....(1)

Consider the horizontal direction. Take the direction to the left as positive.

$T \sin \theta = ma$ .....(2)

(2) ÷ (1),

$$\tan \theta = \frac{a}{g}$$

$$\tan 10° = \frac{a}{9.81}$$

$$a = 1.73 \text{ m s}^{-2}$$

The car's acceleration is 1.73 m s^{-2}.

Think about...

How will θ change if an air freshener of a larger mass is used?

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 In each of the following cases, determine whether the vertical and horizontal components of the net force acting on the object is zero. Neglect air resistance.

	An object being hung by a rope	A patient being pulled up by a helicopter and moving with a constant velocity	A man wearing a jetpack and speeding up in horizontal direction only
Vertical component	zero / non-zero	zero / non-zero	zero / non-zero
Horizontal component	zero / non-zero	zero / non-zero	zero / non-zero

Learn by Practice 6

1 Figure a shows the top view of a 500-kg boat. The boat is being pulled by two forces in the direction shown and keeps moving along the x–direction with acceleration.

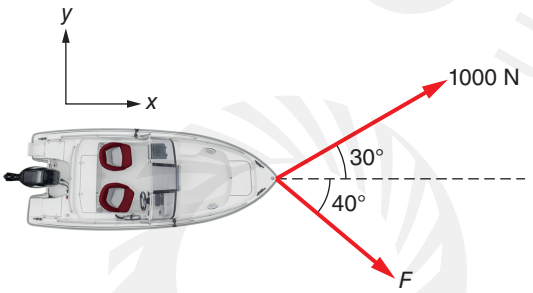


Fig a

(a) What is the net force on the boat along the y-direction?

(b) Hence, find F.

(c) Find the acceleration of the boat.

2 A block of mass 3 kg is pulled by a force F on a smooth horizontal plane (Fig b) and accelerates horizontally to the right at 0.5 m s^{-2}.

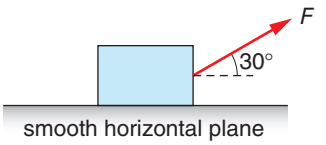


Fig b

(a) Find F.

(b) Find the normal force acting on the block by the plane.

3 An object of mass 200 kg is hung by a cable from a helicopter (Fig c). When the helicopter moves horizontally to the right with a constant acceleration, the cable remains at 80° to the horizontal.

(a) Find the tension in the string.

(b) Find the acceleration of the object.

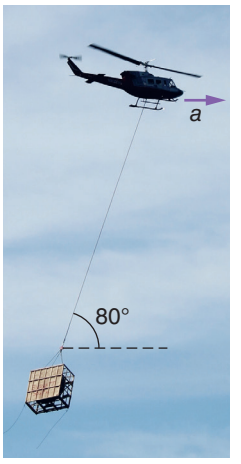


Fig c

3 Objects moving on an inclined plane

Apart from the horizontal and vertical components, a force can be resolved into any other two perpendicular components. Usually, we set one of the components parallel to the moving direction (or the direction of acceleration) of the object. In this part, we will focus on objects moving on an inclined plane and learn how to resolve forces in these cases.

Example 12 Component of weight along a slope

Patrick of mass 70 kg skis down a slope that is inclined at 20° to the horizontal. Find the component of his weight along the slope.

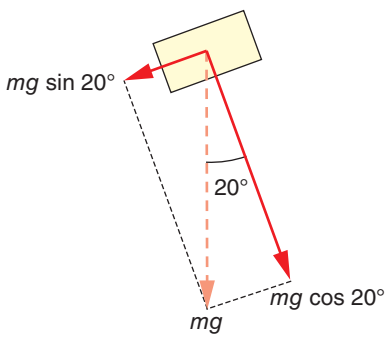
Solution

Component of weight along slope

$= mg \sin 20°$

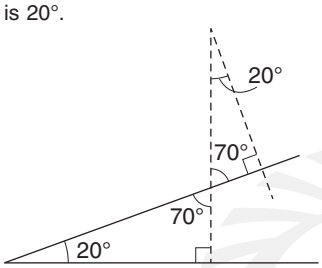
$= 70 \times 9.81 \times \sin 20°$

$= 235$ N



Tip

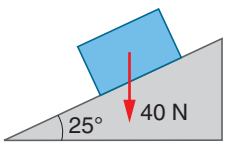
The following figure may help you understand why the angle is 20°.



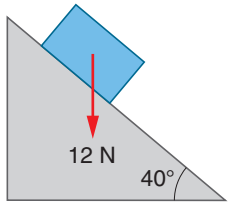
Learn by practice 7

1 Resolve the weight of each of the following objects into components parallel and perpendicular to the inclined plane. Then find the magnitudes of each component.

(a)



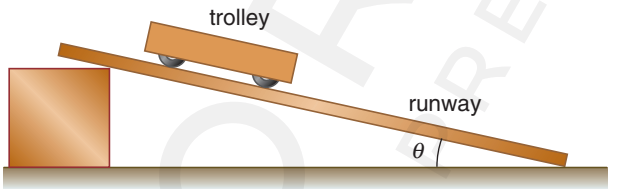
(b)



Resolving forces along the moving direction allows us to find the resultant force in this direction. This helps us analyse the motion along this direction.

Example 13 Friction-compensated runway

When studying motion, we often set up a friction-compensated runway to eliminate the effect of friction. It is inclined in such a way that when a stationary trolley on it is given a push down the runway, the trolley runs down at a constant velocity.



Suppose a friction-compensated runway is inclined at 20°. A trolley of mass 0.8 kg runs down the runway at a constant velocity after being given a push.

(a) Draw a free-body diagram for the trolley.

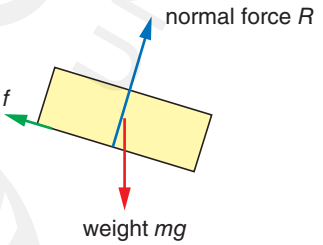
(b) Find the magnitude of the friction f and the normal force R acting on the trolley.

(c) Will the trolley still move with a constant velocity after being given a harder push down the runway? Explain briefly.

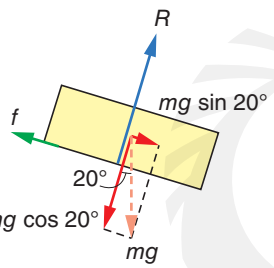
(d) Will the trolley still move with a constant velocity after being given a push up the runway? Explain briefly.

Solution

(a)

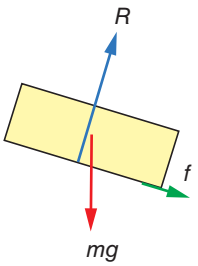


Tip



(b) Consider the direction along the runway.
$f = mg \sin 20° = 0.8 \times 9.81 \times \sin 20° = 2.68$ N
Consider the direction perpendicular to the runway.
$R = mg \cos 20° = 0.8 \times 9.81 \times \cos 20° = 7.37$ N

Tip



(c) Yes, this is because the net force acting on it is zero after the pushing force disappears.

(d) No. When the trolley moves up the runway, both the friction and the component of its weight point down the runway. Therefore, there is a net force pointing down the runway and the trolley should not move with a constant velocity.

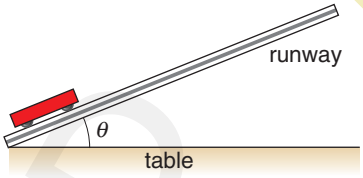
☆ HKDSE 2015 Paper 1B Q4

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | A trolley moving along an inclined runway

A trolley of mass 2 kg is given a sharp push up a runway inclined at θ to a horizontal table. It moves up the runway and then slides back down. The magnitudes of its accelerations when moving upwards and downwards are 0.5 m s^{-2} and 0.3 m s^{-2} respectively.

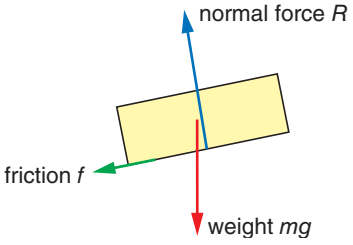


(a) Draw the free-body diagrams for the trolley during the upward journey and the downward journey. (3 marks)

(b) Find the magnitude of friction f acting on the trolley. f has the same value when the trolley moves up and down the runway. (3 marks)

Solution

(a) During the upward journey,

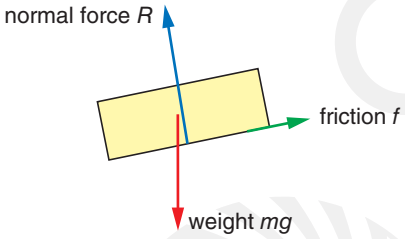


Common mistake
Some students wrongly include the resultant force or the weight component $mg \sin \theta$ in the diagram.

(Correct friction) 1A

(All other forces correct) 1A

During the downward journey,



(Correct friction) 1A

(b) Take the direction down the runway as positive.

Consider the upward journey (with acceleration a_1). By $F_{net} = ma$,

$mg \sin \theta + f = ma_1$ .....(1) 1M

Consider the downward journey (with acceleration a_2). By $F_{net} = ma$,

$mg \sin \theta - f = ma_2$ .....(2) 1M

Common mistake
Students may not be able to set up two equations as the trolley moves up and down the runway, or they may fail to solve the equations for f.

(1) − (2),

$2f = ma_1 - ma_2$

$2f = 2 \times 0.5 - 2 \times 0.3$

$f = 0.2$ N 1A

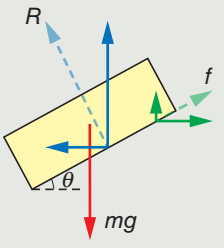
▶ Revision 4 Q37 (p.193)

Know more

Components parallel and perpendicular to an inclined plane

As mentioned before, when we resolve a force into two perpendicular components, one of the components should be parallel to the **direction of acceleration** of the object. Do you know why?

Consider an object moving down a rough inclined plane. If we resolve forces into horizontal and vertical components, we will need to handle the horizontal and vertical components of acceleration as well. However, if we resolve forces into components parallel and perpendicular to the direction of acceleration, the equations will be much simpler.

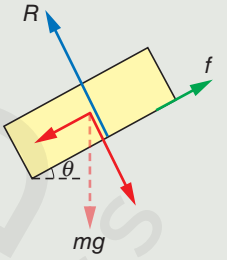


Vertical direction:

$mg + (-f \sin\theta) + (-R \cos\theta) = m \times (a \sin\theta)$

Horizontal direction:

$R \sin\theta + (-f \cos\theta) = m \times (a \cos\theta)$



Perpendicular to acceleration:

$R + (-mg \cos\theta) = 0$

Parallel to acceleration:

$mg \sin\theta + (-f) = ma$

Learn by Practice 8

1 A 4-kg block moves down a rough plane inclined at 30° to the horizontal (Fig a).

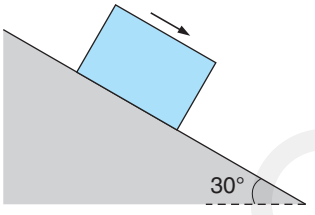


Fig a

Find the magnitude of the friction and that of the normal force acting on the block by the plane if the block moves

(a) at a constant velocity of 0.2 m s^{-1},

(b) at a constant acceleration of 0.2 m s^{-2}.

2 A 5-kg block is pushed up by a force along a smooth plane inclined at 40° to the horizontal (Fig b). If the acceleration of the block is 3 m s^{-2} up the plane, find the magnitude of the pushing force.

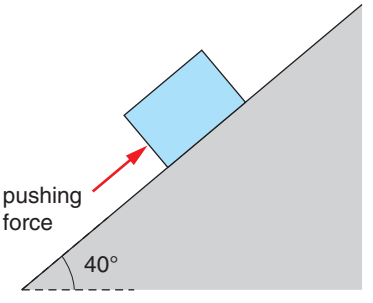


Fig b

Exercise 4.3

★**1** A block of mass 3 kg is at rest on a rough inclined plane (Fig a). A 20-N force perpendicular to the plane acts on the block.

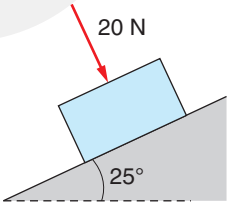


Fig a

What is the normal force acting on the block by the plane?

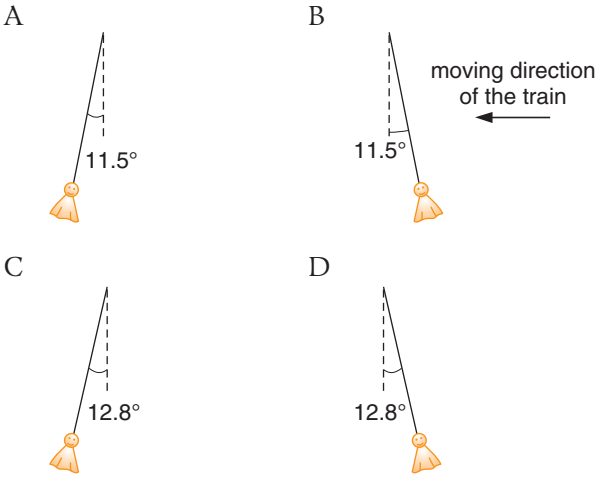
A 6.67 N

B 20 N

C 32.4 N

D 46.7 N

*2 An ornament is hung inside a train. When the train moving towards the left slows down at 2 m s^{-2}, which of the following figures best represents the position of the ornament?



3 To remove a car from a ditch, the driver ties a rope tightly between the car and a tree. When the driver pulls at the centre of the rope with a force of 500 N (Fig b), the car and the rope remain at rest. Find the tension in the rope.

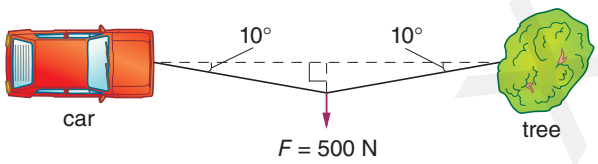


Fig b

4 A woman of mass 55 kg is pulled by a rope and accelerates horizontally to the left (Fig c). The tension in the rope is 100 N and makes an angle of 25° to the horizontal. The resistive force acting on her is 50 N to the right. Find her acceleration.

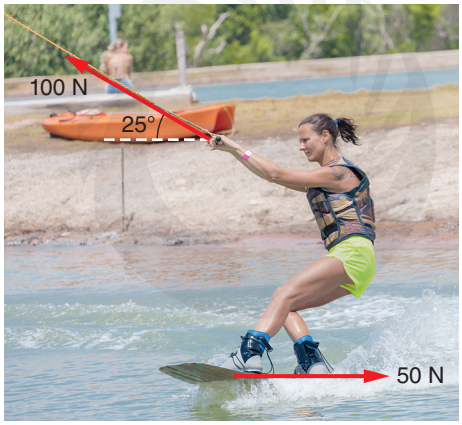


Fig c

*5 The V-22 Osprey is an aircraft that can take off and land vertically. This is because the orientation of the rotors can be changed (Fig d). Suppose the total mass of the aircraft with load is 20 000 kg.

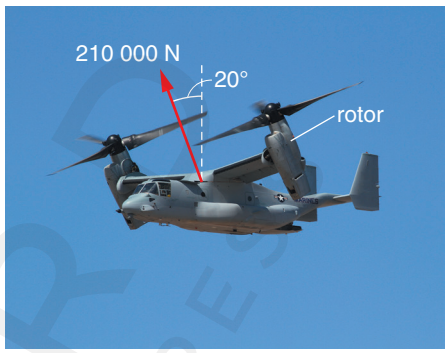


Fig d

(a) At a certain instant, the rotors are tilted to produce an upward force of 210 000 N that acts at an angle of 20° to the vertical (Fig d). Find the acceleration of the aircraft.

(b) During landing, the rotors point vertically upwards. If the aircraft slows down at 1 m s^{-2} when it is moving vertically downwards, what is the force produced by the rotors?

*6 A man pushes a trolley up a slope (Fig e). His pushing force is 200 N and is parallel to the slope. The slope is at an angle of 10° to the horizontal. The friction acting on the trolley is 80 N. Consider the trolley and its contents as a single object of mass 50 kg.

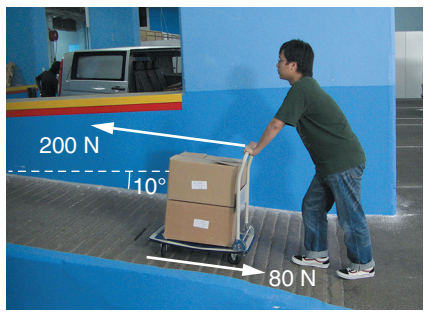


Fig e

(a) Find the normal force acting on the trolley by the slope.

(b) Find the acceleration of the trolley.

(c) The man releases the trolley for a short time. Find the acceleration of the trolley just after it is released.

Q36 Exam Report: This question was on force and acceleration. Candidates were required to comprehend a short passage on accelerometers as well as their use in smart phones. Candidates' performance was far from satisfactory.

(a) Only the more able ones drew the free-body diagram correctly. Most failed to explain clearly that a net upward force was required when the box accelerated upward and thus the tension would be greater (than the weight) and the spring extended further (*h* decreased).

(b) Candidates did poorly in this part. Not many obtained the correct reading of the spring balance using the information given. Quite a number of them did not understand what quantity is being measured by a spring balance and wrongly gave their answers in metres. Few candidates found the acceleration correctly using the equation $T - W = ma$. Some wrongly included mass $m = 5$ kg in their calculation.

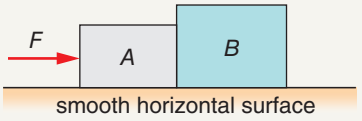
(c) In this part, only the most able ones deduced the display mode correctly by considering the angle rotated by the phone or the component of the acceleration due to gravity along the *y*–axis of the phone. Many candidates were weak in resolving vectors and some were confused by the negative sign associated with the data.

Key concepts 4

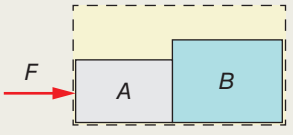
1 System of objects

Objects moving as a whole (with the same velocity and the same acceleration) can be treated as a system

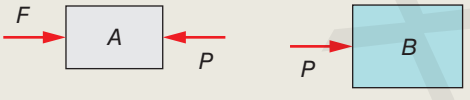
- Objects with their sides in contact



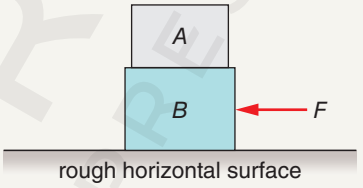
Treated as a system



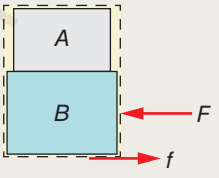
Treated separately



- Stacked objects



Treated as a system



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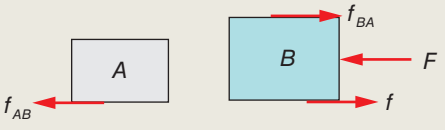
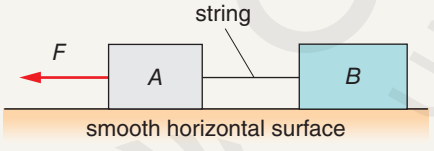
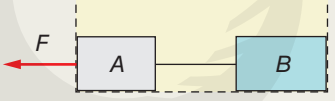


- Connected objects

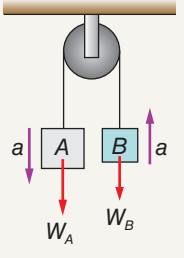


Treated as a system



Treated separately

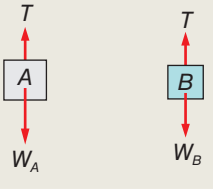
- Objects hanging over a pulley



Treated as a system

$W_A - W_B = (m_A + m_B)a$

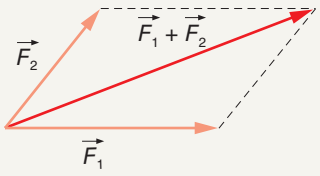
Treated separately



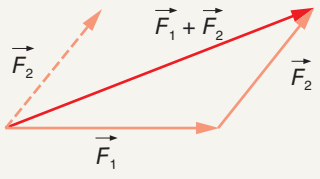
2 Addition and resolution of forces

- Graphical methods:

  Parallelogram of forces method



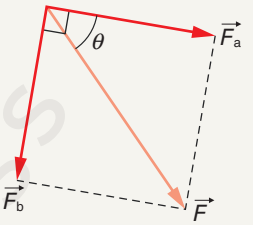
  Tip-to-tail method



- A force $\vec{F}$ can be resolved into two **perpendicular** components $\vec{F}_a$ and $\vec{F}_b$.

  $F_a = F \cos \theta$

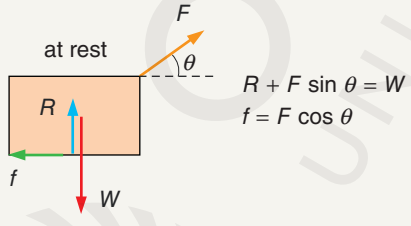
  $F_b = F \sin \theta$

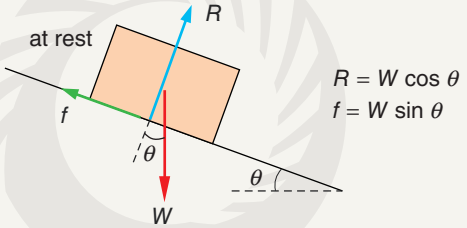


- Coplanar forces can be added by first resolving them.

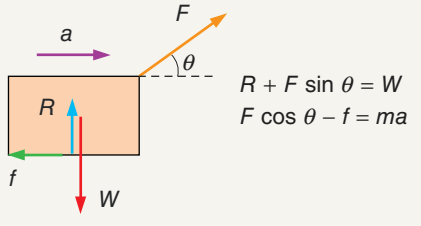
3 Applying Newton's laws of motion

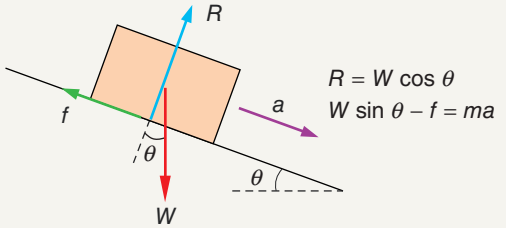
- For an object at rest or moving at constant velocity, by Newton's first law,





- For an object moving with acceleration a, by Newton's second law,





(a) A few candidates failed to describe correctly the motion, which was based on the velocity–time graph given.

(b) (i) Most candidates worked out the correct acceleration using the graph but some overlooked calculating the magnitude.

(ii) This part was well answered.

Key terms

1 component 分量 p.167 **2** resolve 分解 p.167

(c) Many candidates were able to draw a free-body diagram showing the forces (with labels) acting on the block as it moved up the plane. However, some candidates wrongly included the resultant force and/or the weight component along the plane ($mg \sin \theta$) in addition to mg. They also wrongly drew the direction and line of action of the frictional force.

(d) This part was poorly answered. Only the most able candidates were able to set up correctly two equations for the forces acting on the block as it decelerated up and accelerated down the inclined plane.

Revision 4

Multiple-choice questions

★1 4.2 The magnitude of the resultant of a 3-N force and a 4-N force cannot be

A 0 N.

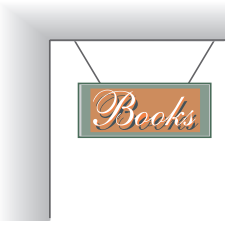
B 1 N.

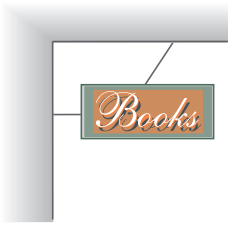
C 4 N.

D 7 N.

★2 4.3 A signboard is hung by two identical wires. Which of the following is the safest way to hang it?

A

B 

C 

D

★3 4.1 Blocks *X*, *Y* and *Z* are put together on a smooth horizontal plane and a horizontal force *F* acts on *X* (Fig a).

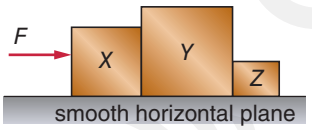


Fig a

Which of the following statements is/are correct?

(1) The force acting on *Z* by *Y* is equal to *F*.

(2) The force acting on *Z* by *Y* has the same magnitude as the force acting on *X* by *Y*.

(3) The force acting on *Y* by *X* has the same magnitude as the force acting on *X* by *Y*.

A (2) only

B (3) only

C (1) and (2) only

D (1) and (3) only

★4 4.1 A box is placed on the floor of a container truck such that it is up against the wall of the container (Fig b). The truck is moving towards the right at a constant velocity.

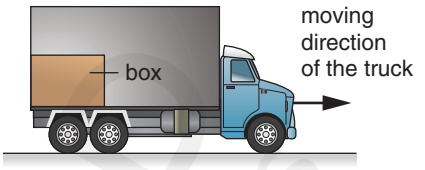
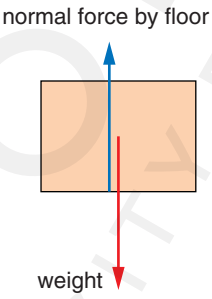
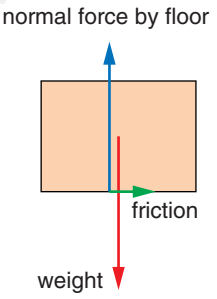
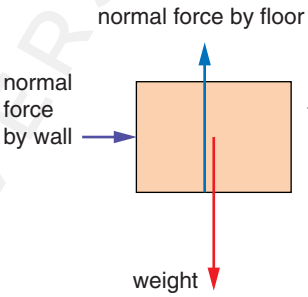


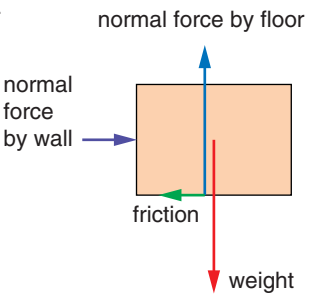
Fig b

Which of the following is the correct free-body diagram for the box?

A 

B 

C 

D 

★5 4.2 A kangaroo of mass 80 kg lands on the ground (Fig c). At this moment, the force acting on it by the ground is 960 N. This force makes an angle of 75° to the horizontal. Find the magnitude of the net force acting on it.

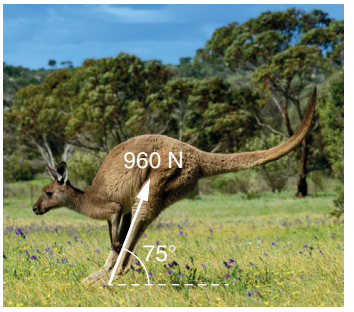


Fig c

A 175 N

B 286 N

C 1070 N

D 1730 N

★6 4.3 A trolley travels down an inclined plane from rest. Its v–t graph is shown in Figure d.

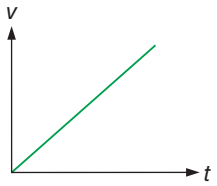
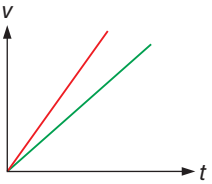
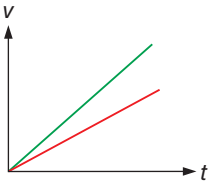
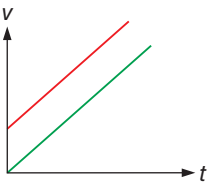


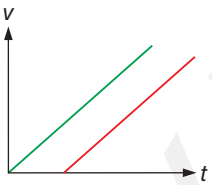
Fig d

Then the inclined angle of the plane is decreased and the trolley travels down the plane from rest again. Which of the following best shows the new v–t graph (the red line) of the trolley?

A 

B 

C 

D 

★7 Misc Three blocks are hung over two smooth pulleys as shown (Fig e). If block W is at rest, find the tension T in the string holding it.

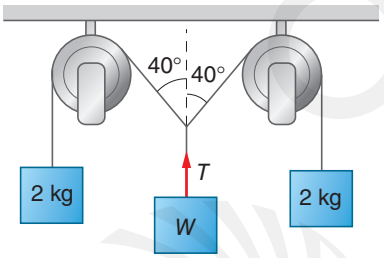


Fig e

A 25.2 N
B 30.1 N
C 32.9 N
D 39.2 N

★8 4.1 Identical blocks P and Q are connected by a string and placed on a smooth horizontal plane (Fig f). A 5-N force pulls P horizontally towards the left while a 7-N force pulls Q horizontally towards the right. What is the net force acting on P? Take the direction to the right as positive.

Fig f

A −2 N
B −1 N
C 1 N
D 2 N

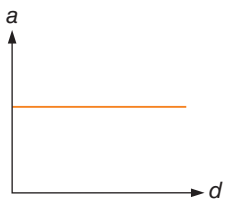
★★9 Misc A wolf of mass 30 kg is at rest on the slanted surface of a stone (Fig g). This surface makes an angle of 20° to the horizontal. The stone remains stationary on horizontal ground. What is the friction acting on the stone by the ground?

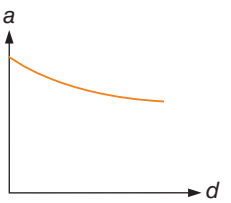
Fig g

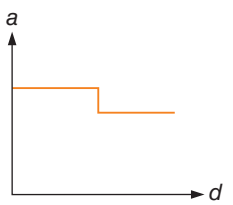
A 0 N
B 94.6 N
C 101 N
D 277 N

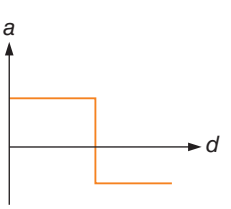
Refer p.155

★★10 4.3 A trolley is given a sharp push so that it moves up a rough inclined plane. It moves back down the plane after reaching its highest position. Which of the following graphs best illustrates how the acceleration a of the trolley varies with the distance travelled d?

A 

B 

C 

D 

Refer p.179

**11 Misc Block X of mass 2 kg is placed on a rough inclined plane while block Y of mass 0.5 kg is placed on block X (Fig h). Both blocks are stationary. The rough plane is at an angle of 15° to the horizontal.

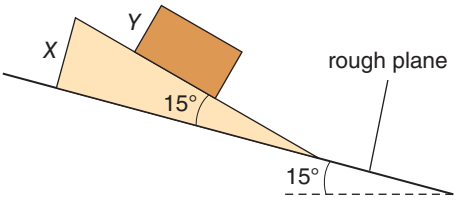


Fig h

Find the magnitude of force acting on X by the rough plane.

A 19.0 N B 19.6 N
C 23.7 N D 24.5 N

Refer p.154

**12 4.3 A man of mass 70 kg sits in a train that accelerates constantly at 0.8 m s^{-2} up a rail inclined at 15° (Fig i). What is the magnitude of the force acting on the man by the train?

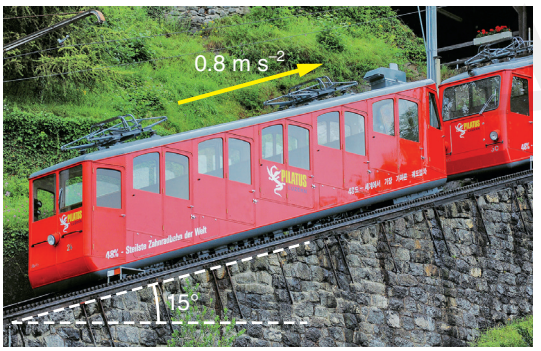


Fig i

A 216 N B 687 N
C 703 N D 743 N

Refer p.178

**13 Misc Blocks X and Y are placed on a smooth plane inclined at 30° to the horizontal. A horizontal force of 50 N acts on X (Fig j). The weight of X is 20 N and the weight of Y is 30 N. Find the magnitude of the force acting on X by Y.

A 0
B 7.32 N
C 26.0 N
D 36.6 N

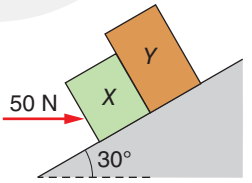


Fig j

Refer p.156

**14 Misc A spring gun of mass 1 kg projects a ball of mass 0.2 kg vertically upwards. At a certain instant, the acceleration of the ball is 5 m s^{-2} upwards (Fig k). What is the normal force acting on the spring gun by the floor at this instant?

A 9.81 N
B 10.8 N
C 11.8 N
D 12.8 N

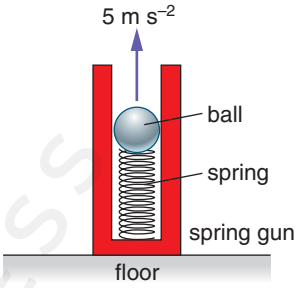


Fig k

Refer p.156

15 HKDSE 2012 Paper 1A Q8 4.1

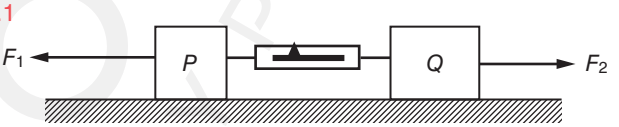


Fig l

Blocks P and Q of mass m and $2m$ respectively are connected by a light spring balance and placed on a smooth horizontal surface as shown. If horizontal forces F_1 and F_2 (with $F_1 > F_2$) act on P and Q respectively and the whole system moves to the left with constant acceleration, what is the reading of the spring balance?

A $\frac{2F_1 - F_2}{3}$ B $\frac{2(F_1 - F_2)}{3}$
C $\frac{2F_1 + F_2}{3}$ (36%) D $\frac{F_1 + 2F_2}{3}$

16 HKDSE 2013 Paper 1A Q5 4.3

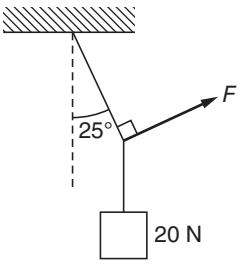


Fig m

A block of weight 20 N is suspended by a light string from the ceiling. A force F is applied such that the block is displaced to one side with the string making an angle of 25° with the vertical as shown. Find the magnitude of F.

A 8.5 N (44%) B 9.3 N
C 18.1 N D 47.3 N

17 HKDSE 2018 Paper 1A Q5
4.3

A block of weight 8 N is suspended from a horizontal ceiling by light inextensible strings to two different points P and Q as shown. The strings are equal in length.

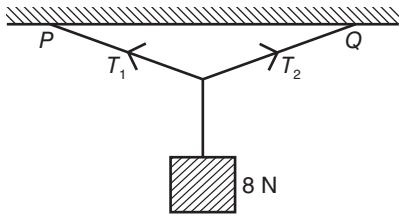


Fig n

Which of the following descriptions about the tensions T_1 and T_2 in the two strings is/are correct?

(1) The magnitude of T_1 must be greater than 4 N.

(2) The maximum value of T_2 would not exceed 8 N.

(3) The resultant force of T_1 and T_2 is zero.

A (1) only (38%)

B (3) only (10%)

C (1) and (2) only (28%)

D (2) and (3) only (24%)

18 HKDSE 2018 Paper 1A Q8
4.3

A block of mass m is placed on a smooth incline making an angle θ with the horizontal as shown. When a force of magnitude F parallel to the incline is applied to the block, it travels up the incline with an acceleration a. If the applied force becomes $2F$, what would the magnitude of the acceleration be?

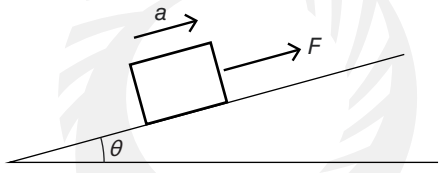


Fig o

A greater than $2a$ (37%)

B equal to $2a$ (26%)

C between a and $2a$ (26%)

D whether it is greater than $2a$, equal to $2a$ or between a and $2a$ depends on the value of θ (11%)

Q19 Exam Report: Almost one-third of the candidates chose option D only considering the 5 kg block as a free falling object.

19 HKDSE 2019 Paper 1A Q12
4.1

Two blocks of respective masses 2 kg and 5 kg are connected by a light inextensible string which passes over a smooth fixed light pulley as shown. The system is released from rest when the 5-kg block is 3 m above the ground. What is the speed of the 5-kg block just when reaching the ground? Neglect air resistance. (g = 9.81 m s^{-2})

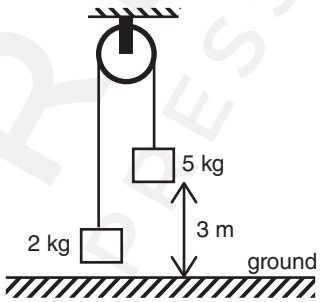


Fig p

A 5.0 m s^{-1} (34%)

B 6.0 m s^{-1} (20%)

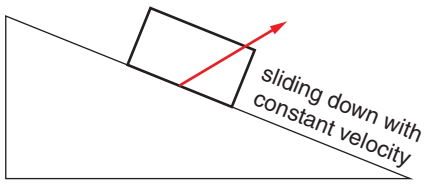
C 6.5 m s^{-1} (15%)

D 7.7 m s^{-1} (31%)

20 HKDSE 2021 Paper 1A Q5
Misc

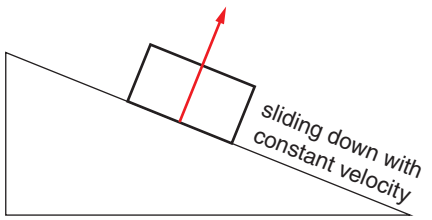
A block is sliding down a rough incline with constant velocity as shown. Which arrow indicates the direction of **the resultant force acting on the block by the incline**? Neglect air resistance.

A



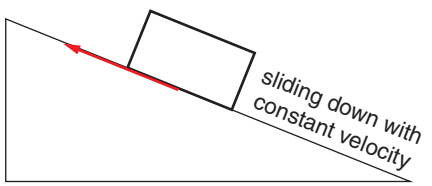
(13%)

B



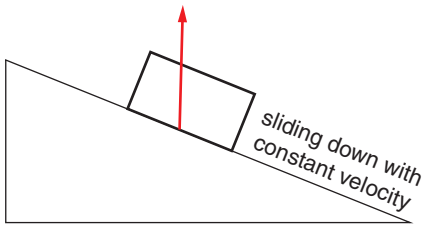
(29%)

C



(23%)

D



(35%)

Q20 Exam Report: Just over one-third of the candidates realized that the resultant force from the incline acting on the block in uniform motion should balance the block's own weight.

Conventional questions

★21 4.1 Balls *X* and *Y* have masses of 200 g and 50 g respectively. They are hung by strings *P* and *Q* and remain stationary (Fig q).

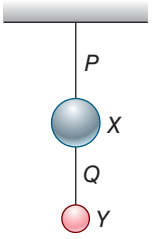


Fig q

(a) Draw a free-body diagram for each ball. (2 marks)

(b) Find the tension in each string. (3 marks)

(c) Will the tensions in the strings change if the balls are hung as shown in Figure r? (2 marks)

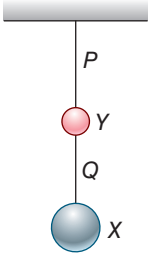


Fig r

(d) How will the tensions in the strings change if the balls in Figure q are hung in a lift that is accelerating downwards? (2 marks)

★22 4.1 A toy train consists of two parts *X* and *Y* which are connected by a string. *X* is pulled by a 0.6-N force to the right on a smooth horizontal surface (Fig s). The masses of *X* and *Y* are 200 g and 100 g respectively.

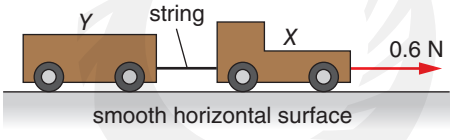


Fig s

(a) Find the acceleration of the train. (2 marks)

(b) Find the tension in the string. (2 marks)

(c) Ivy says that *Y* will stop if the string breaks suddenly. Comment on her statement. (2 marks)

★23 4.3 A man is standing at the mid-point *X* of a cable (Fig t). The tension in the cable is 1280 N.

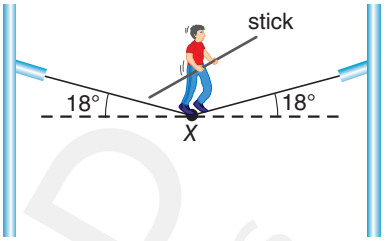


Fig t

(a) Find the resultant force of the tensions acting on point *X*. (3 marks)

(b) Find the total mass of the man and the stick. (2 marks)

★24 4.3 A boy slides down a sand dune (Fig u). The total mass of him and the sledge is 50 kg. The dune surface is inclined at 50° to the horizontal. Consider the boy and the sledge as a single object.

Fig u

(a) Draw a free-body diagram for the boy. (2 marks)

(b) What is the direction of the net force acting on him if he

(i) slides down faster and faster along a straight path;

(ii) slides down slowly at a constant speed along a straight path;

(iii) stays at rest on the dune? (3 marks)

(c) What is his acceleration if the friction between him and the dune is 300 N? (2 marks)

★**25** 4.3 A robot flies by pushing air with force F_1 through its back and force F_2 through its legs (Fig v). F_1 makes an angle of 30° with the vertical while F_2 points vertically downwards.

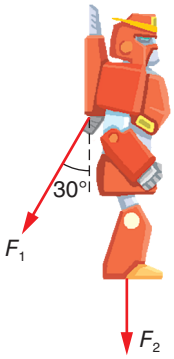


Fig v

(a) Explain why the robot can fly by pushing air downwards. (3 marks)

(b) Describe how the robot can remain stationary in the air. (2 marks)

(c) The robot is now accelerating at 5 m s^{-2} horizontally. The mass of the robot is 10 000 kg.

(i) Find F_1 and F_2. (4 marks)

(ii) How will F_1 and F_2 change if it moves horizontally with a higher acceleration? (2 marks)

★**26** 4.3 A man of mass 70 kg is doing exercises. He pulls a cable which is connected to a weight stack of mass 50 kg over a smooth pulley (Fig w). The man and the weight stack hung by the cable remain at rest.

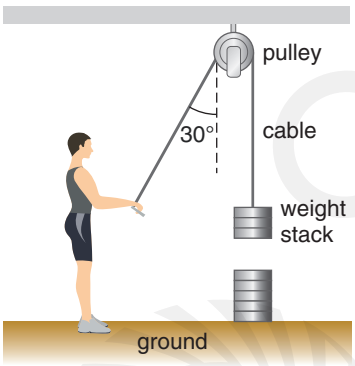


Fig w

(a) Find the tension in the cable. (2 marks)

(b) Find the normal force acting on the man by the ground. (2 marks)

(c) Can the man stand still if the ground is smooth? Explain briefly. (2 marks)

(d) Find the magnitude of the force that the man exerts on the ground. (2 marks)

(e) Then the man pulls the cable with the same force while the angle between the cable and the vertical increases. How does the friction acting on him by the ground change? (1 mark)

★**27** 4.3 Figure x shows a helicopter moving horizontally to the left. U is the upward force acting on it. U points to the right at angle of 15° to the vertical. The only forces acting on the helicopter are U and the weight.

Fig x

(a) Describe the motion of the helicopter at the instant shown. (1 mark)

(b) If U is 50 000 N, what is the mass of the helicopter? (2 marks)

(c) In order to make the helicopter move at a constant velocity, Alan suggests adjusting the direction of U to vertically upwards. Comment on his suggestion. (2 marks)

★**28** 4.3 Figure y shows a crane lifting a steel tube. The mass of the tube is 450 kg. Each of the four ropes connected to the tube makes an angle of 12° to the vertical. The tensions in the four ropes are always equal.

Fig y

(a) The steel tube is held stationary.

(i) What is the resultant force acting on it by the ropes? (2 marks)

(ii) Find the tension in each rope. (2 marks)

(b) The steel tube is then raised upwards with an acceleration of 0.3 m s^{-2}. Find the tension in each rope in this case. (3 marks)

(c) A rope breaks if the tension in it is too great. Is it safer to hang the steel tube on shorter ropes? Explain briefly. (3 marks)

★29 4.1 Block X of mass 2 kg on a rough horizontal table is connected to block Y of mass m with a light inextensible string over a smooth pulley (Fig z). The blocks are released at time $t = 0$ and Y reaches the ground at $t = 5$ s. Figure aa shows how the velocity v of X varies with t. The direction to the right is taken as positive.

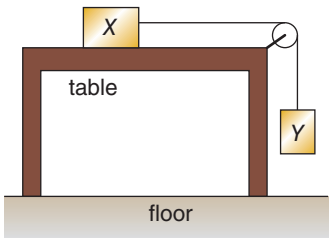


Fig z

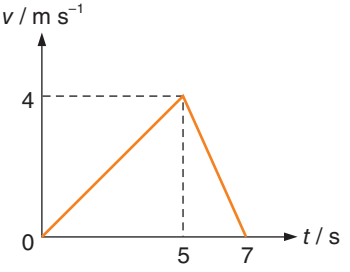


Fig aa

(a) Describe the motion of X during $t = 0$ to $t = 7$ s. (2 marks)

(b) Find the friction between X and the table. (3 marks)

(c) Find m. (2 marks)

(d) Sketch how v changes with t in Figure aa if Y is replaced by a more massive block. (2 marks)

★30 Misc The Peak Tram is a famous railway in Hong Kong. There are two tramcars in the system. Each tramcar is pulled by a cable to move up and down the Peak (Fig ab). The cables are wound on the same drum. When the drum is driven by a motor to rotate, one tramcar is pulled up while the other moves down with the same speed.

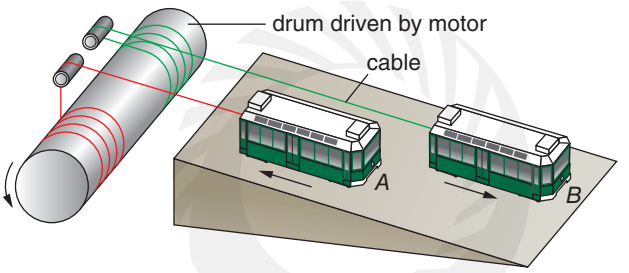


Fig ab

Suppose tramcars A and B have the same mass of 20 000 kg. The friction acting on each tramcar when it moves is 10 000 N. The slope is inclined at 20° to the horizontal. Figure ac shows the v–t graph of A. The direction up the slope is taken as positive.

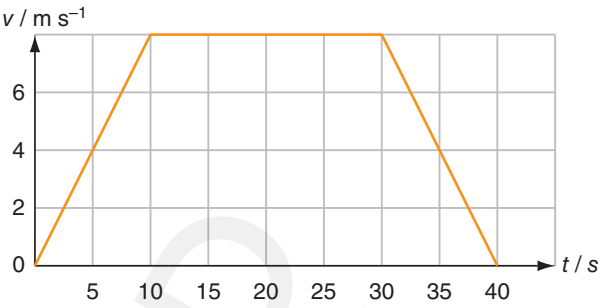


Fig ac

(a) Find the acceleration of A at $t = 5$ s. (2 marks)

(b) Find the tension in the cable pulling A at $t = 5$ s. (2 marks)

(c) Find the tension in the cable pulling B at $t = 5$ s. (2 marks)

(d) Find the tension in the cable pulling A at $t = 35$ s. (2 marks)

★31 4.1 Block X is placed on top of block Y and block Y is placed on a rough horizontal plane (Fig ad). The mass of X is 2 kg and the mass of Y is 3 kg. If the blocks move on the plane, the friction acting on Y by the plane is 5 N. The blocks are at rest initially. A horizontal force F acts on Y.

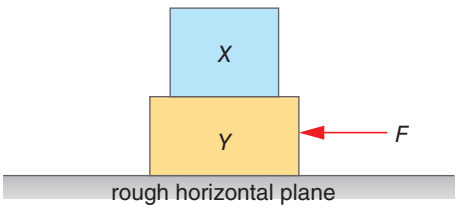


Fig ad

(a) What is the friction acting on X by Y if $F = 3$ N? (1 mark)

(b) Then F is increased to 10 N and the two blocks move together without slipping.

(i) Find the friction acting on X by Y. (3 marks)

(ii) Find the friction acting on Y by X. (1 mark)

(c) The maximum friction between X and Y is 15 N. Find the minimum value of F to make X slide on Y. (3 marks)

*32 Misc Two identical smooth blocks X and Y are placed together on a smooth horizontal plane (Fig ae). The mass of each block is 2 kg. A 10-N force acts horizontally on X and the blocks move together towards the right.

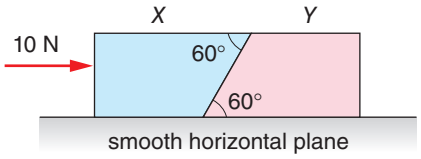


Fig ae

(a) Find the acceleration of X. (2 marks)

(b) Find the magnitude of the force acting on X by Y. (2 marks)

(c) Find the normal force acting on Y by the plane. (3 marks)

**33 4.3 An engineer designs a lift that takes passengers up a slope along a straight line (Fig af). The floor of the lift is kept horizontal.

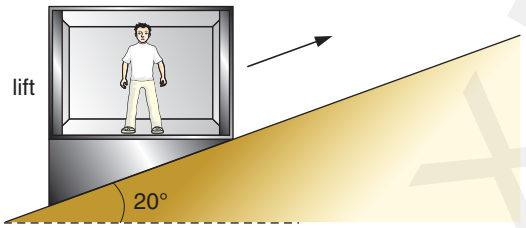


Fig af

Suppose a man of mass 65 kg takes the lift up the slope. The force acting on the man by the lift can be resolved into two perpendicular components F_1 and F_2, with F_1 parallel to the slope and F_2 perpendicular to the slope. Find the magnitude of F_1 and F_2 when the lift

(a) moves at a constant velocity; (3 marks)

(b) decelerates at 0.8 m s^{-2}. (3 marks)

Refer p.174

**34 Misc A 800-g box is connected to a 500-g block by an inextensible string running over a smooth pulley (Fig ag).

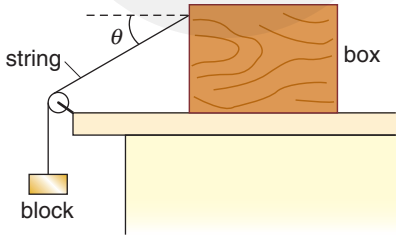


Fig ag

When the block is released, the box is pulled towards the left. The friction between the box and the table is 3.5 N. After speeding up for a certain time, the box starts to slow down.

(a) Why does the box slow down? (1 mark)

(b) Find θ at the instant that the box starts to slow down. (3 marks)

Refer p.174

**35 Misc A 5-kg block X is placed on a smooth inclined plane as shown (Fig ah). The plane makes an angle θ to the horizontal.

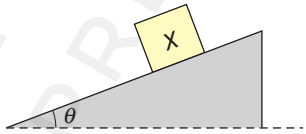


Fig ah

(a) X is released from rest and its v–t graph is as shown in Figure ai.

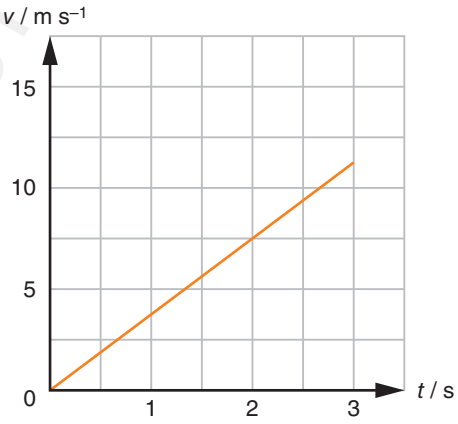


Fig ai

(i) Find the acceleration of X. (2 marks)

(ii) Find θ. (2 marks)

(b) X is now kept stationary on the inclined plane by a horizontal force F as shown (Fig aj).

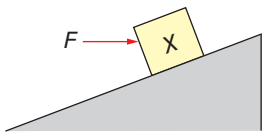


Fig aj

(i) Find F. (2 marks)

(ii) Find the magnitude of the force exerted by X on the inclined plane. (2 marks)

(iii) If the inclined plane is rough, explain whether the minimum value of F required to hold X stationary becomes larger or smaller. (3 marks)

(c) The force F is now removed and X is connected to a 3-kg block Y by a string over a smooth pulley (Fig ak). Find the acceleration of X when the blocks are released. (2 marks)

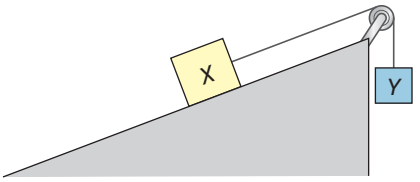


Fig ak

Refer p.161

36 HKDSE 2014 Paper 1B Q4

Misc

Read the following description about **accelerometers** and answer the questions that follow.

An accelerometer is a device for measuring acceleration. The following example illustrates the principle of a simple accelerometer. An object of mass M is suspended by a spring balance inside a box. If the box is at rest without acceleration, the object is h above the bottom of the box (Fig al). When the box accelerates upwards, h decreases (Fig am). Likewise, when the box accelerates downwards, h increases (Fig an). Since it is known that the tension of the spring balance is directly proportional to its extension, we can therefore determine the magnitude and direction of the box's acceleration by measuring h.

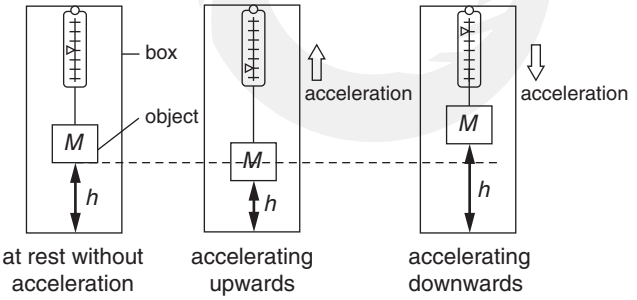


Fig al *Fig am* *Fig an*

(a) Draw a labelled free-body diagram in the space provided below showing the forces acting on the object when the box accelerates upwards with acceleration a. Explain why h decreases in this case. (4 marks)

Fig ao

(b) The scale of the spring balance is calibrated such that the pointer moves 1 cm for a change of 2 N of force. The weight of the object is 5 N. If h decreases by 0.5 cm compared to the situation in Figure al, what is the reading of the balance? Hence find the magnitude of the corresponding acceleration of the box. (acceleration due to gravity $g = 9.81\ \text{m s}^{-2}$) (3 marks)

Electronic accelerometers employing similar principles are widely used in smart phones. The phone's orientation is detected by a set of built-in accelerometers each detecting the acceleration due to gravity along mutually perpendicular axes of the phone. A phone in vertical orientation is shown on the left of Figure ap, its accelerometer along the Y axis would be sensing the acceleration due to gravity, denoted by $a_Y = -g$. When the phone is rotated about a horizontal axis perpendicular to both X and Y axes by more than 45°, 'portrait display' would change to 'landscape display' as shown on the right of Figure ap.

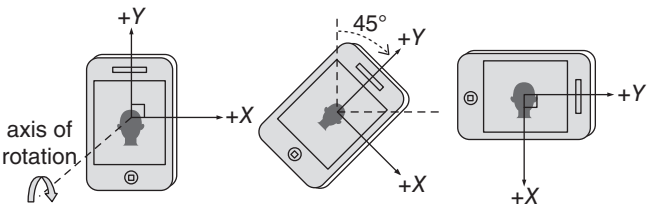


Fig ap

(c) What would the kind of display be if the phone is rotated clockwise until the acceleration a_Y sensed by the accelerometer along the rotated Y axis is $-0.5g$? Explain. (2 marks)

37 **HKDSE 2015 Paper 1B Q4**
4.3

The motion of a block on an inclined plane can be investigated using a motion sensor connected to a computer (not shown in Figure aq).

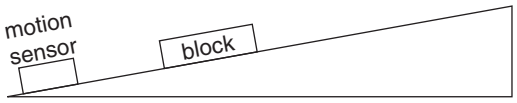


Fig aq

A block is given a push up a rough inclined plane and then released. The velocity–time (v–t) graph recorded by the sensor is shown below. Assume that the frictional force acting on the block is constant in magnitude throughout the motion. Neglect air resistance. (g = 9.81 m s^{-2})

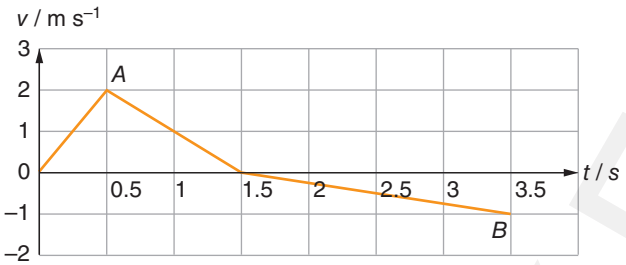


Fig ar

Point A on the graph corresponds to the instant at which the push is removed.

(a) Describe the block's motion **from *A* to *B***. (2 marks)

(b) (i) Find the magnitude of the block's acceleration from t = 1.5 s to t = 3.5 s. (2 marks)

(ii) Draw the corresponding acceleration–time (a–t) graph of the block. With the direction up the inclined plane taken to be positive, the part during which the block is being pushed has been drawn for you. (2 marks)

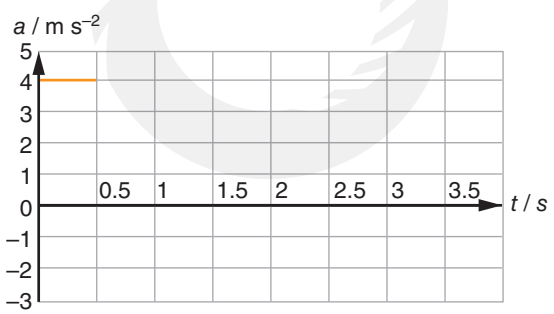


Fig as

(c) Draw a free-body diagram to show the force(s) (with labels) acting on the block as it moves up the inclined plane after the push is removed. (2 marks)

Fig at

(d) If the mass of the block is 1.0 kg, find the magnitude of the frictional force. (3 marks)

38 **HKDSE 2015 Paper 1B Q5**
4.3

You are given a long light string, a protractor and a metal ball with a hook.

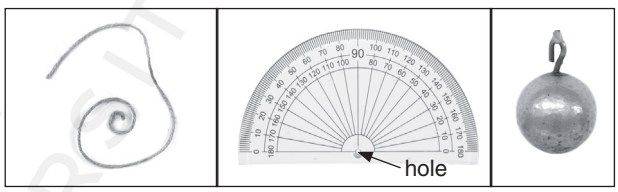


Fig au

Suppose you are inside a train which is at rest initially and later it travels along a straight horizontal track with constant acceleration. With the aid of a diagram, describe how to measure the acceleration of the train. Show your working including mathematical derivation. (6 marks)

39 **HKDSE 2020 Paper 1B Q5**
4.3

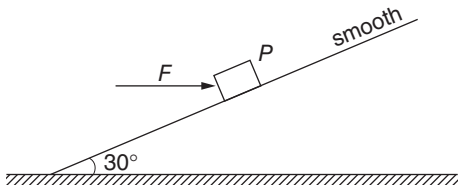


Fig av

(a) A block P of mass 10 kg is kept stationary on a smooth incline by a horizontal force F as shown in Figure av. The incline makes an angle of 30° with the horizontal. (g = 9.81 m s^{-2})

(i) On Figure av, indicate and label all other forces acting on P. (2 marks)

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(ii) Find the magnitudes of the force F and the force exerted by the block on the incline respectively. (3 marks)

(b) Now F is removed and neglect air resistance.

(i) What is the magnitude of the acceleration of the block? (1 mark)

(ii) Explain whether the force exerted by the block on the incline would increase, decrease or remain unchanged when compared with that in (a)(ii). (2 marks)

40 HKDSE 2021 Paper 1B Q4

4.1

A trolley is connected a hanger of mass 20 g by a light inextensible string as shown in Figure aw. Four slotted weights, each of mass 20 g, are loaded onto the trolley. The experiment is designed to investigate the relationship between the net force acting on the system (trolley and slotted weights with hanger) and its acceleration. The acceleration a is measured after the trolley is released on the smooth horizontal runway.

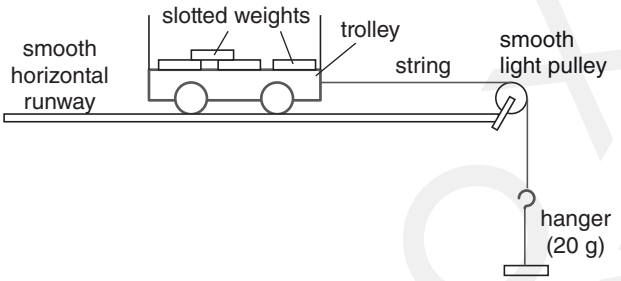

Fig aw

The experiment is repeated by transferring the slotted weights one by one from the trolley to the hanger so as to increase the mass hanging, m.

no. of weights transferred to the hanger	0	1	2	3	4
mass hanging m / kg	0.020	0.040	0.060	0.080	0.100

Table a

The results obtained are used for plotting a graph of a against m as shown in Figure ax. Neglect both air resistance and the frictional forces acting on the trolley. (g = 9.81 m s^{-2})

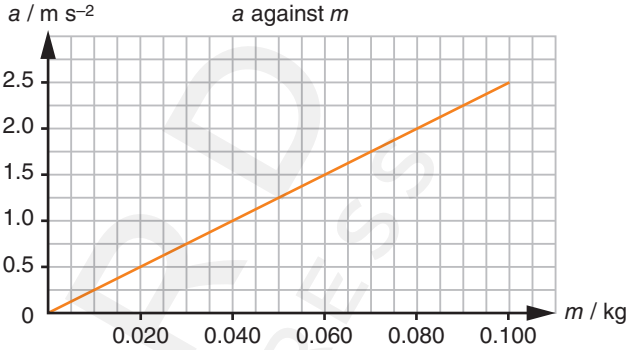

Fig ax

(a) (i) After the trolley is released, indicate in the figures below (1) the horizontal force(s) acting on the loaded trolley, and (2) the force(s) acting on the hanger. (2 marks)

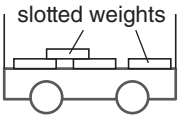

Fig ay

Fig az

(ii) Is the tension in the string equal to, greater than or smaller than the weight of the mass hanging when the system is released? Explain. (2 marks)

(iii) By considering the motion of the whole system, or otherwise, write an equation relating m, a and mass M of the trolley. (1 mark)

(b) Calculate the slope of the graph. Hence find M using the result of (a)(iii). (3 marks)

(a) (i) Most were able to indicated the forces acting on the block when it was treated as a free body. However, a few candidates made mistakes in either the direction or the labelling of the normal force.

(ii) Candidates' performance was poor. Not many managed to resolve the forces and correctly set up the equations required. Some wrongly recalled $R = mg \cos\theta$ for the normal force in such a situation that involved an extra force F.

(b) (i) A lot of the candidates answered this part correctly though some of them used F (the force being removed) to find the acceleration.

(ii) This part was unfamiliar to candidates and some failed to realize that the perpendicular component of F had a part to play in such a situation.